asset-class-return-correlations-part-1

November 11, 2026

0.1 Post Updates

Update 11/1/2026: Split this post into two parts. This post now covers the correlations and rolling correlations, and the total correlation metric, decile analysis, and trading strategy have moved to part 2.

0.2 Introduction

In this post, we'll take a look at historical return correlations between asset classes, including rolling correlations and how those correlations have changed over time. Since first publishing this post, we've split it into two parts, and part 2 uses these correlations to build a market timing signal and a trading strategy.

Python Imports

[1]:
```
import os
import sys
import warnings

from itertools import combinations, count
from pathlib import Path

# Data Handling
import pandas as pd

# Display
from IPython.display import Markdown, display

# Suppress warnings
warnings.filterwarnings("ignore")

# Add src/ to the path so settings.py and the helper modules can be imported
src_directory = str(Path.cwd().parents[2] / "src")
if src_directory not in sys.path:
    sys.path.append(src_directory)

from settings import config

DATA_DIR = config("DATA_DIR")  # keep only the directories this post uses
```

0.3 Python Functions

Here are the functions needed for this project:

- load_data: Load data from a CSV, Excel, or Pickle file into a pandas DataFrame.
- pandas_set_decimal_places: Set the number of decimal places displayed for floating-point numbers in pandas.
- plot_heatmap: Plot a heatmap from a DataFrame, such as a correlation matrix, with annotated cells and a configurable color scale.
- plot_time_series: Plot the timeseries data from a DataFrame for a specified date range and columns.
- yf_pull_data: Download daily price data from Yahoo Finance and export it.

[2]:
```
from load_data import load_data
from pandas_set_decimal_places import pandas_set_decimal_places
from plot_heatmap import plot_heatmap
from plot_time_series import plot_time_series
from yf_pull_data import yf_pull_data
```

[3]:
```
pandas_set_decimal_places(2)
figure_number = count(1)
table_number = count(1)
```

0.4 Data Overview

For this exercise, we will (mostly) use ETFs as a proxy for asset classes and will use the following:

- Large Cap US Stocks / S&P 500 – IVV (iShares S&P 500 ETF)
- Mid Cap US Stocks / S&P MidCap 400 – IJH (iShares S&P MidCap 400 ETF)
- Small Cap US Stocks / S&P SmallCap 600 – IJR (iShares S&P SmallCap 600 ETF)
- US Tech Stocks / Nasdaq 100 – QQQ (Invesco QQQ Trust, Series 1)
- Large & Mid Cap US Stocks – IWB (iShares Russell 1000 ETF)
- Small Cap US Stocks – IWM (iShares Russell 2000 ETF)
- US Value Stocks – IWD (iShares Russell 1000 Value ETF)
- US Growth Stocks – IWF (iShares Russell 1000 Growth ETF)
- International Developed Market Stocks – EFA (iShares MSCI EAFE ETF)
- Emerging Market Stocks – EEM (iShares MSCI Emerging Markets ETF)
- European Stocks – IEV (iShares S&P Europe 350 ETF)
- Short-Term US Treasuries – SHY (iShares 1-3 Year Treasury Bond ETF)
- Medium-Term US Treasuries – IEF (iShares 7-10 Year Treasury Bond ETF)
- Long-Term US Treasuries – TLT (iShares 20+ Year Treasury Bond ETF)
- Aggregate Bonds – AGG (iShares Core U.S. Aggregate Bond ETF)
- Gold – GLD (SPDR Gold Shares)
- Commodities – GSG (iShares S&P GSCI Commodity-Indexed Trust)
- Real Estate – IYR (iShares U.S. Real Estate ETF)
- Bitcoin – BTC-USD (Bitcoin USD)
- Ethereum – ETH-USD (Ethereum USD)

For all of these, we will use the adjusted closing price, which accounts for dividends and stock splits.

0.4.1 Acquire & Plot Data

We'll now pull the data for all the ETFs and cryptocurrencies listed above.

[4]:
```
# Create list of tickers to pull data for
us_equity_tickers = ["IVV", "IJH", "IJR", "QQQ", "IWB", "IWD", "IWF", "IWM"]
intl_equity_tickers = ["EFA", "EEM", "IEV"]
equity_tickers = us_equity_tickers + intl_equity_tickers
bond_tickers = ["SHY", "IEF", "TLT", "AGG"]
commodity_tickers = ["GLD", "GSG"]
real_estate_tickers = ["IYR"]
cryptoasset_tickers = ["BTC-USD", "ETH-USD"]

etf_tickers = (
    us_equity_tickers
    + intl_equity_tickers
    + bond_tickers
    + commodity_tickers
    + real_estate_tickers
)

tickers_dict = {
    "IVV": "Large Cap US Stocks / S&P 500 -- IVV (iShares S&P 500 ETF)",
    "IJH": "Mid Cap US Stocks / S&P MidCap 400 -- IJH (iShares S&P MidCap 400␣
 ↪ETF)",
    "IJR": "Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap␣
 ↪600 ETF)",
    "QQQ": "US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1)",
    "IWB": "Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF)",
    "IWM": "Small Cap US Stocks -- IWM (iShares Russell 2000 ETF)",
    "IWD": "Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value␣
 ↪ETF)",
    "IWF": "Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000␣
 ↪Growth ETF)",
    "EFA": "International Developed Market Stocks -- EFA (iShares MSCI EAFE␣
 ↪ETF)",
    "EEM": "Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF)",
    "IEV": "European Stocks -- IEV (iShares S&P Europe 350 ETF)",
    "SHY": "Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond␣
 ↪ETF)",
    "IEF": "Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond␣
 ↪ETF)",
    "TLT": "Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond␣
 ↪ETF)",
    "AGG": "Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF)",
    "GLD": "Gold -- GLD (SPDR Gold Shares)",
    "GSG": "Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust)",
```

```
    "IYR": "Real Estate -- IYR (iShares U.S. Real Estate ETF)",
    "BTC-USD": "Bitcoin -- BTC-USD (Bitcoin USD)",
    "ETH-USD": "Ethereum -- ETH-USD (Ethereum USD)",
}
```

[5]:
```
for ticker in etf_tickers:
    yf_pull_data(
        base_directory=DATA_DIR,
        ticker=ticker,
        adjusted=False,
        source="Yahoo_Finance",
        asset_class="Exchange_Traded_Funds",
        excel_export=True,
        pickle_export=True,
        output_confirmation=False,
    )

for ticker in cryptoasset_tickers:
    yf_pull_data(
        base_directory=DATA_DIR,
        ticker=ticker,
        adjusted=False,
        source="Yahoo_Finance",
        asset_class="Cryptocurrencies",
        excel_export=True,
        pickle_export=True,
        output_confirmation=False,
    )
```

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  data.sort_index(level=0, axis=1, inplace=True)

/home/jared/Cloud_Storage/Dropbox/Websites/jaredszajkowski.github.io/.venv/lib/p
ython3.14/site-packages/yfinance/utils.py:568: Pandas4Warning: The inplace
keyword in DataFrame.sort_index is deprecated and will be removed in a future
version (PDEP-8).
See https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-inplace
for more details.
  quotes.sort_index(inplace=True)
/home/jared/Cloud_Storage/Dropbox/Websites/jaredszajkowski.github.io/.venv/lib/p
ython3.14/site-packages/yfinance/multi.py:220: Pandas4Warning: The inplace
keyword in DataFrame.rename is deprecated and will be removed in a future
version (PDEP-8).
See https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-inplace
for more details.
  data.rename(columns=ctx.isins, inplace=True)
/home/jared/Cloud_Storage/Dropbox/Websites/jaredszajkowski.github.io/.venv/lib/p
ython3.14/site-packages/yfinance/multi.py:224: Pandas4Warning: The inplace
keyword in DataFrame.sort_index is deprecated and will be removed in a future
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See https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-inplace
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  data.sort_index(level=0, axis=1, inplace=True)

/home/jared/Cloud_Storage/Dropbox/Websites/jaredszajkowski.github.io/.venv/lib/p
ython3.14/site-packages/yfinance/utils.py:568: Pandas4Warning: The inplace
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See https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-inplace
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ython3.14/site-packages/yfinance/multi.py:220: Pandas4Warning: The inplace
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See https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-inplace
for more details.
  data.rename(columns=ctx.isins, inplace=True)
/home/jared/Cloud_Storage/Dropbox/Websites/jaredszajkowski.github.io/.venv/lib/p
ython3.14/site-packages/yfinance/multi.py:224: Pandas4Warning: The inplace
keyword in DataFrame.sort_index is deprecated and will be removed in a future
version (PDEP-8).
See https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-inplace
for more details.
```

```
  data.sort_index(level=0, axis=1, inplace=True)

/home/jared/Cloud_Storage/Dropbox/Websites/jaredszajkowski.github.io/.venv/lib/p
ython3.14/site-packages/yfinance/utils.py:568: Pandas4Warning: The inplace
keyword in DataFrame.sort_index is deprecated and will be removed in a future
version (PDEP-8).
See https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-inplace
for more details.
  quotes.sort_index(inplace=True)
/home/jared/Cloud_Storage/Dropbox/Websites/jaredszajkowski.github.io/.venv/lib/p
ython3.14/site-packages/yfinance/utils.py:585: Pandas4Warning: The inplace
keyword in DataFrame.set_index is deprecated and will be removed in a future
version (PDEP-8).
See https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-inplace
for more details.
  dividends.set_index("date", inplace=True)
/home/jared/Cloud_Storage/Dropbox/Websites/jaredszajkowski.github.io/.venv/lib/p
ython3.14/site-packages/yfinance/utils.py:587: Pandas4Warning: The inplace
keyword in DataFrame.sort_index is deprecated and will be removed in a future
version (PDEP-8).
See https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-inplace
for more details.
  dividends.sort_index(inplace=True)
/home/jared/Cloud_Storage/Dropbox/Websites/jaredszajkowski.github.io/.venv/lib/p
ython3.14/site-packages/yfinance/utils.py:604: Pandas4Warning: The inplace
keyword in DataFrame.set_index is deprecated and will be removed in a future
version (PDEP-8).
See https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-inplace
for more details.
  splits.set_index("date", inplace=True)
/home/jared/Cloud_Storage/Dropbox/Websites/jaredszajkowski.github.io/.venv/lib/p
ython3.14/site-packages/yfinance/utils.py:606: Pandas4Warning: The inplace
keyword in DataFrame.sort_index is deprecated and will be removed in a future
version (PDEP-8).
See https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-inplace
for more details.
  splits.sort_index(inplace=True)
/home/jared/Cloud_Storage/Dropbox/Websites/jaredszajkowski.github.io/.venv/lib/p
ython3.14/site-packages/yfinance/multi.py:220: Pandas4Warning: The inplace
keyword in DataFrame.rename is deprecated and will be removed in a future
version (PDEP-8).
See https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-inplace
for more details.
  data.rename(columns=ctx.isins, inplace=True)
/home/jared/Cloud_Storage/Dropbox/Websites/jaredszajkowski.github.io/.venv/lib/p
ython3.14/site-packages/yfinance/multi.py:224: Pandas4Warning: The inplace
keyword in DataFrame.sort_index is deprecated and will be removed in a future
version (PDEP-8).
```

```
See https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-inplace
for more details.
  data.sort_index(level=0, axis=1, inplace=True)

/home/jared/Cloud_Storage/Dropbox/Websites/jaredszajkowski.github.io/.venv/lib/p
ython3.14/site-packages/yfinance/utils.py:568: Pandas4Warning: The inplace
keyword in DataFrame.sort_index is deprecated and will be removed in a future
version (PDEP-8).
See https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-inplace
for more details.
  quotes.sort_index(inplace=True)
/home/jared/Cloud_Storage/Dropbox/Websites/jaredszajkowski.github.io/.venv/lib/p
ython3.14/site-packages/yfinance/multi.py:220: Pandas4Warning: The inplace
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version (PDEP-8).
See https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-inplace
for more details.
  data.rename(columns=ctx.isins, inplace=True)
/home/jared/Cloud_Storage/Dropbox/Websites/jaredszajkowski.github.io/.venv/lib/p
ython3.14/site-packages/yfinance/multi.py:224: Pandas4Warning: The inplace
keyword in DataFrame.sort_index is deprecated and will be removed in a future
version (PDEP-8).
See https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-inplace
for more details.
  data.sort_index(level=0, axis=1, inplace=True)

/home/jared/Cloud_Storage/Dropbox/Websites/jaredszajkowski.github.io/.venv/lib/p
ython3.14/site-packages/yfinance/utils.py:568: Pandas4Warning: The inplace
keyword in DataFrame.sort_index is deprecated and will be removed in a future
version (PDEP-8).
See https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-inplace
for more details.
  quotes.sort_index(inplace=True)
/home/jared/Cloud_Storage/Dropbox/Websites/jaredszajkowski.github.io/.venv/lib/p
ython3.14/site-packages/yfinance/multi.py:220: Pandas4Warning: The inplace
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version (PDEP-8).
See https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-inplace
for more details.
  data.rename(columns=ctx.isins, inplace=True)
/home/jared/Cloud_Storage/Dropbox/Websites/jaredszajkowski.github.io/.venv/lib/p
ython3.14/site-packages/yfinance/multi.py:224: Pandas4Warning: The inplace
keyword in DataFrame.sort_index is deprecated and will be removed in a future
version (PDEP-8).
See https://pandas.pydata.org/docs/dev/whatsnew/v3.1.0.html#deprecation-inplace
for more details.
  data.sort_index(level=0, axis=1, inplace=True)
```

0.4.2 Load & Filter Data

We'll then peform the following: * Load data * Rename columns to include the ticker (e.g. "QQQ_Close", "QQQ_Adj_Close", etc.) * Drop all columns except for "Adj Close" * Calculate the daily returns * Combine the data into a single DataFrame * Check the inception date for each of thge funds

[6]:
```
fund_data = pd.DataFrame()

for ticker in etf_tickers:
    data = load_data(
        base_directory=DATA_DIR,
        ticker=ticker,
        source="Yahoo_Finance",
        asset_class="Exchange_Traded_Funds",
        timeframe="Daily",
        file_format="pickle",
    )

    # Rename columns to "QQQ_Close", etc.
    data = data.rename(
        columns={
            "Adj Close": f"{ticker}_Adj_Close",
            "Close": f"{ticker}_Close",
            "High": f"{ticker}_High",
            "Low": f"{ticker}_Low",
            "Open": f"{ticker}_Open",
            "Volume": f"{ticker}_Volume",
        }
    )

    # Drop all columns except for the adjusted close price and date index
    data = data[[f"{ticker}_Adj_Close"]]

    # Calculate daily returns and add as new column
    data[f"{ticker}_Daily_Return"] = data[f"{ticker}_Adj_Close"].pct_change()

    # Concatenate the data for this ticker with the main fund_data DataFrame
    fund_data = pd.concat([fund_data, data], axis=1)

for ticker in cryptoasset_tickers:
    data = load_data(
        base_directory=DATA_DIR,
        ticker=ticker,
        source="Yahoo_Finance",
        asset_class="Cryptocurrencies",
        timeframe="Daily",
        file_format="pickle",
```

```
    )

    # Rename columns to "BTC-USD_Close", etc.
    data = data.rename(
        columns={
            "Adj Close": f"{ticker}_Adj_Close",
            "Close": f"{ticker}_Close",
            "High": f"{ticker}_High",
            "Low": f"{ticker}_Low",
            "Open": f"{ticker}_Open",
            "Volume": f"{ticker}_Volume",
        }
    )

    # Drop all columns except for the adjusted close price and date index
    data = data[[f"{ticker}_Adj_Close"]]

    # Calculate daily returns and add as new column
    data[f"{ticker}_Daily_Return"] = data[f"{ticker}_Adj_Close"].pct_change()

    # Concatenate the data for this ticker with the main fund_data DataFrame
    fund_data = pd.concat([fund_data, data], axis=1)

display(Markdown(f"Table {next(table_number)}: Fund Data"))
display(fund_data)
```

Table 1: Fund Data

Date	IVV_Adj_Close	IVV_Daily_Return	IJH_Adj_Close	IJH_Daily_Return	\
1999-03-10	NaN	NaN	NaN	NaN	
1999-03-11	NaN	NaN	NaN	NaN	
1999-03-12	NaN	NaN	NaN	NaN	
1999-03-15	NaN	NaN	NaN	NaN	
1999-03-16	NaN	NaN	NaN	NaN	
…	…	…	…	…	
2026-11-06	738.68	0.00	72.11	0.01	
2026-11-07	NaN	NaN	NaN	NaN	
2026-11-08	NaN	NaN	NaN	NaN	
2026-11-09	756.05	0.02	74.39	0.03	
2026-11-10	753.37	-0.00	74.13	-0.00	

Date	IJR_Adj_Close	IJR_Daily_Return	QQQ_Adj_Close	QQQ_Daily_Return	\
1999-03-10	NaN	NaN	42.98	NaN	
1999-03-11	NaN	NaN	43.19	0.00	
1999-03-12	NaN	NaN	42.14	-0.02	
1999-03-15	NaN	NaN	43.35	0.03	

```
1999-03-16            NaN               NaN          43.72              0.01
…                     …                 …            …                  …
2026-11-06         137.64              0.00        656.10              0.01
2026-11-07            NaN               NaN           NaN               NaN
2026-11-08            NaN               NaN           NaN               NaN
2026-11-09         141.72              0.03        679.35              0.04
2026-11-10         141.19             -0.00        674.76             -0.01

            IWB_Adj_Close  IWB_Daily_Return  …  GLD_Adj_Close  \
Date                                         …
1999-03-10            NaN               NaN  …            NaN
1999-03-11            NaN               NaN  …            NaN
1999-03-12            NaN               NaN  …            NaN
1999-03-15            NaN               NaN  …            NaN
1999-03-16            NaN               NaN  …            NaN
…                     …                 …    …            …
2026-11-06         402.58              0.00  …         372.52
2026-11-07            NaN               NaN  …            NaN
2026-11-08            NaN               NaN  …            NaN
2026-11-09         412.33              0.02  …         379.53
2026-11-10         410.85             -0.00  …         369.36

            GLD_Daily_Return  GSG_Adj_Close  GSG_Daily_Return  IYR_Adj_Close  \
Date
1999-03-10               NaN            NaN               NaN            NaN
1999-03-11               NaN            NaN               NaN            NaN
1999-03-12               NaN            NaN               NaN            NaN
1999-03-15               NaN            NaN               NaN            NaN
1999-03-16               NaN            NaN               NaN            NaN
…                        …              …                 …              …
2026-11-06             -0.01          34.46             -0.00         101.62
2026-11-07               NaN            NaN               NaN            NaN
2026-11-08               NaN            NaN               NaN            NaN
2026-11-09              0.02          33.93             -0.02         103.18
2026-11-10             -0.03          33.67             -0.01         102.98

            IYR_Daily_Return  BTC-USD_Adj_Close  BTC-USD_Daily_Return  \
Date
1999-03-10               NaN                NaN                   NaN
1999-03-11               NaN                NaN                   NaN
1999-03-12               NaN                NaN                   NaN
1999-03-15               NaN                NaN                   NaN
1999-03-16               NaN                NaN                   NaN
…                        …                  …                     …
2026-11-06             -0.01           75746.93                 -0.00
2026-11-07               NaN           75747.58                  0.00
2026-11-08               NaN           74584.65                 -0.02
2026-11-09              0.02           75223.15                  0.01
```

```
2026-11-10             -0.00           73720.12                -0.02

            ETH-USD_Adj_Close  ETH-USD_Daily_Return
Date
1999-03-10                NaN                   NaN
1999-03-11                NaN                   NaN
1999-03-12                NaN                   NaN
1999-03-15                NaN                   NaN
1999-03-16                NaN                   NaN
…                         …                     …
2026-11-06            2235.05                  0.00
2026-11-07            2236.32                  0.00
2026-11-08            2120.98                 -0.05
2026-11-09            2131.50                  0.00
2026-11-10            2050.18                 -0.04

[8344 rows x 40 columns]
```

[7]:
```
for ticker in tickers_dict.keys():
    print(f"First date for {ticker}: {fund_data[f'{ticker}_Daily_Return'].
 ↪first_valid_index()}")
```

```
First date for IVV: 2000-05-22 00:00:00
First date for IJH: 2000-05-30 00:00:00
First date for IJR: 2000-05-30 00:00:00
First date for QQQ: 1999-03-11 00:00:00
First date for IWB: 2000-05-22 00:00:00
First date for IWM: 2000-05-30 00:00:00
First date for IWD: 2000-05-30 00:00:00
First date for IWF: 2000-05-30 00:00:00
First date for EFA: 2001-08-28 00:00:00
First date for EEM: 2003-04-15 00:00:00
First date for IEV: 2000-07-31 00:00:00
First date for SHY: 2002-07-31 00:00:00
First date for IEF: 2002-07-31 00:00:00
First date for TLT: 2002-07-31 00:00:00
First date for AGG: 2003-09-30 00:00:00
First date for GLD: 2004-11-19 00:00:00
First date for GSG: 2006-07-24 00:00:00
First date for IYR: 2000-06-20 00:00:00
First date for BTC-USD: 2014-09-18 00:00:00
First date for ETH-USD: 2017-11-10 00:00:00
```

0.4.3 Plot Data

We'll then plot the time series of the adjusted close prices for each of the assets.

[8]:
```
# Combine the etf_tickers and cryptoasset_tickers lists into a single list of all tickers
all_tickers = etf_tickers + cryptoasset_tickers

for ticker in tickers_dict.keys():
    plot_time_series(
        df=fund_data,
        plot_start_date=None,
        plot_end_date=None,
        plot_columns=[f"{ticker}_Adj_Close"],
        title=f"Figure {next(figure_number)}: {tickers_dict[ticker]} Adjusted Close Price",
        x_label="Date",
        x_format="Year",
        x_tick_spacing=1,
        x_tick_start=None,
        x_tick_rotation=30,
        y_label="Price ($)",
        y_format="Decimal",
        y_format_decimal_places="Auto",
        y_tick_spacing="Auto",
        y_tick_rotation=0,
        caption=f"Data Source: Yahoo Finance, Analysis by the Author",
    )
```

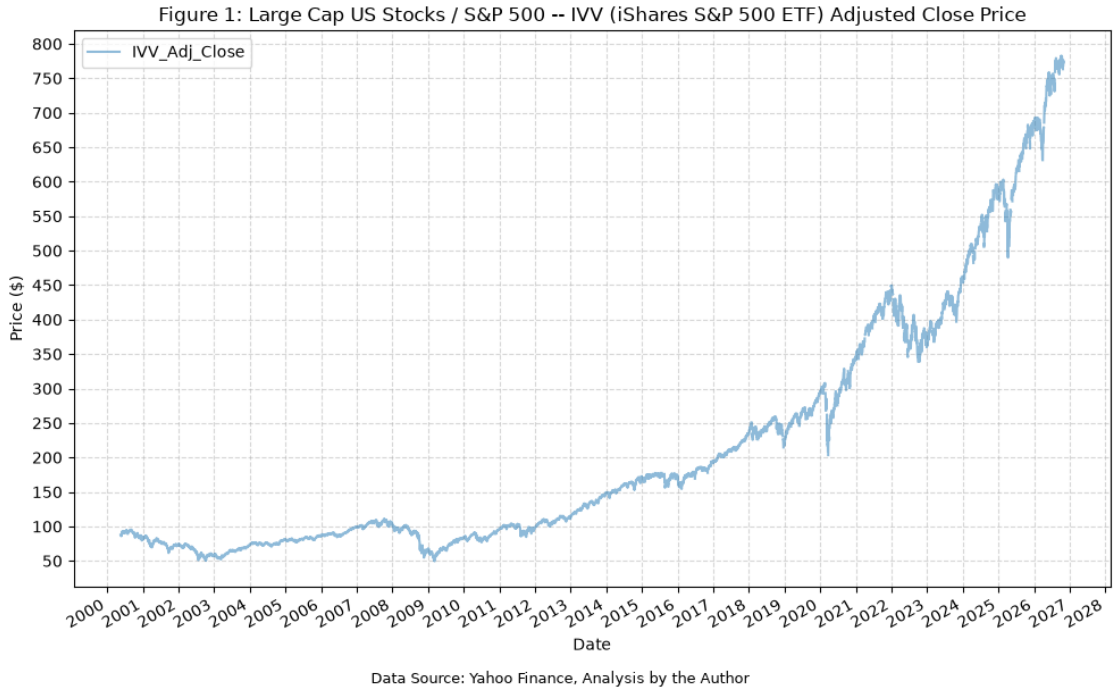
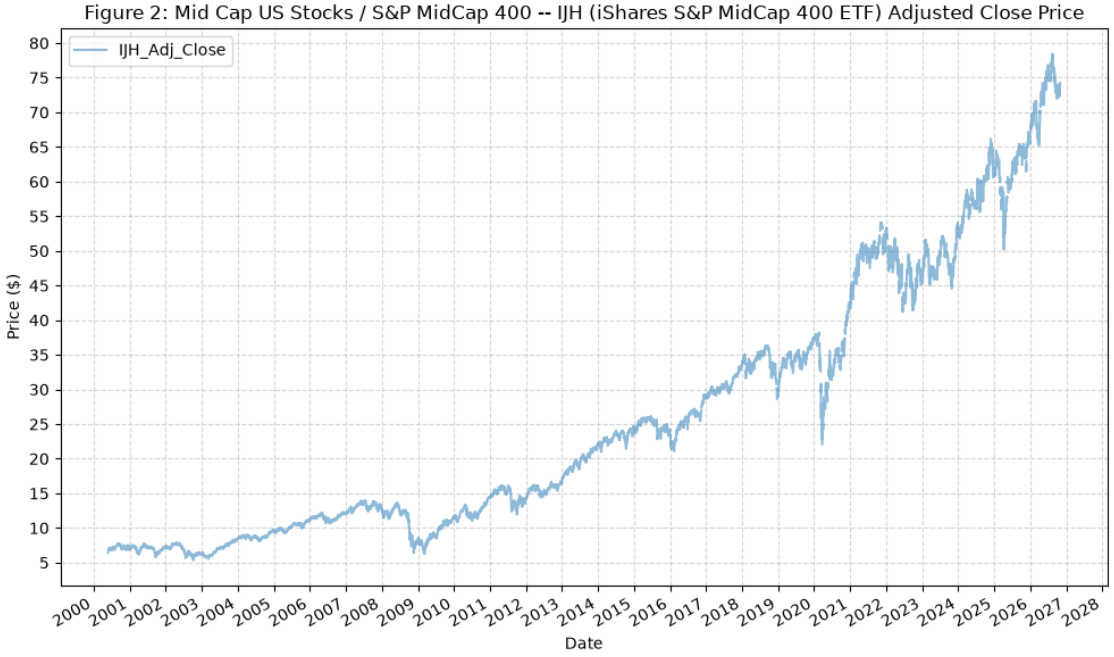
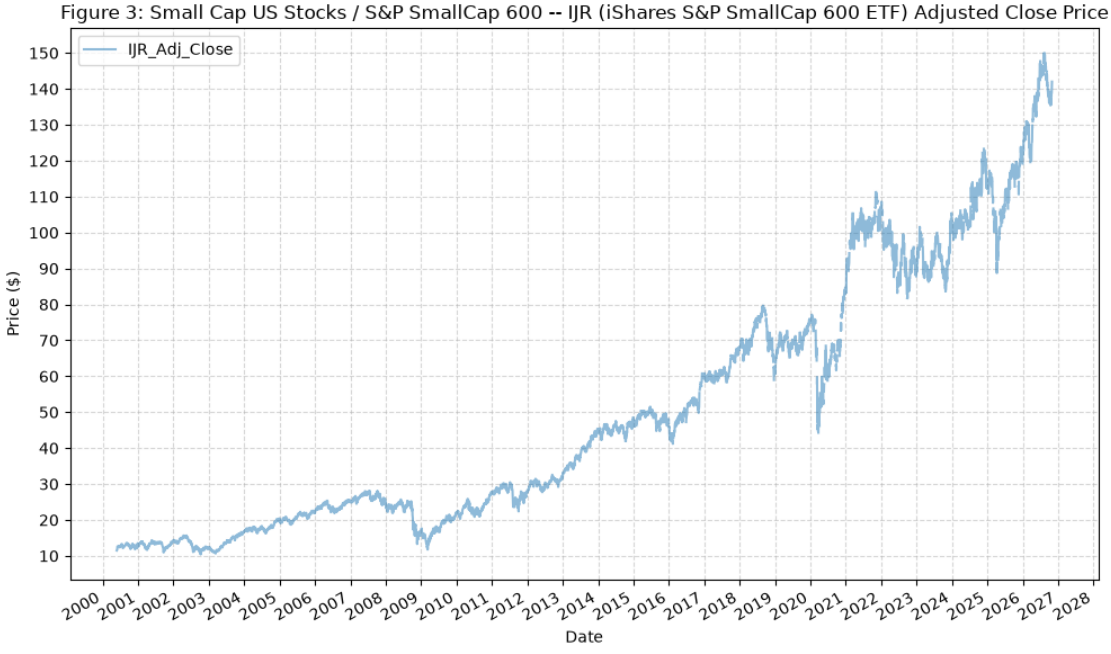


Figure 2: Mid Cap US Stocks / S&P MidCap 400 -- IJH (iShares S&P MidCap 400 ETF) Adjusted Close Price



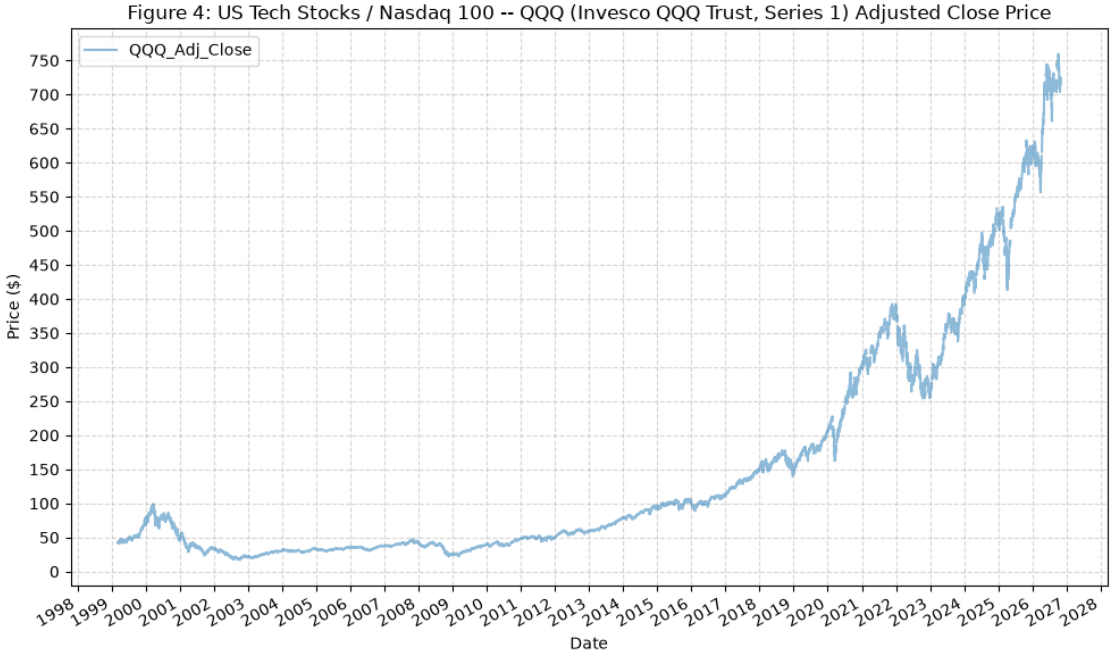
Data Source: Yahoo Finance, Analysis by the Author

Figure 3: Small Cap US Stocks / S&P SmallCap 600 -- IJR (iShares S&P SmallCap 600 ETF) Adjusted Close Price



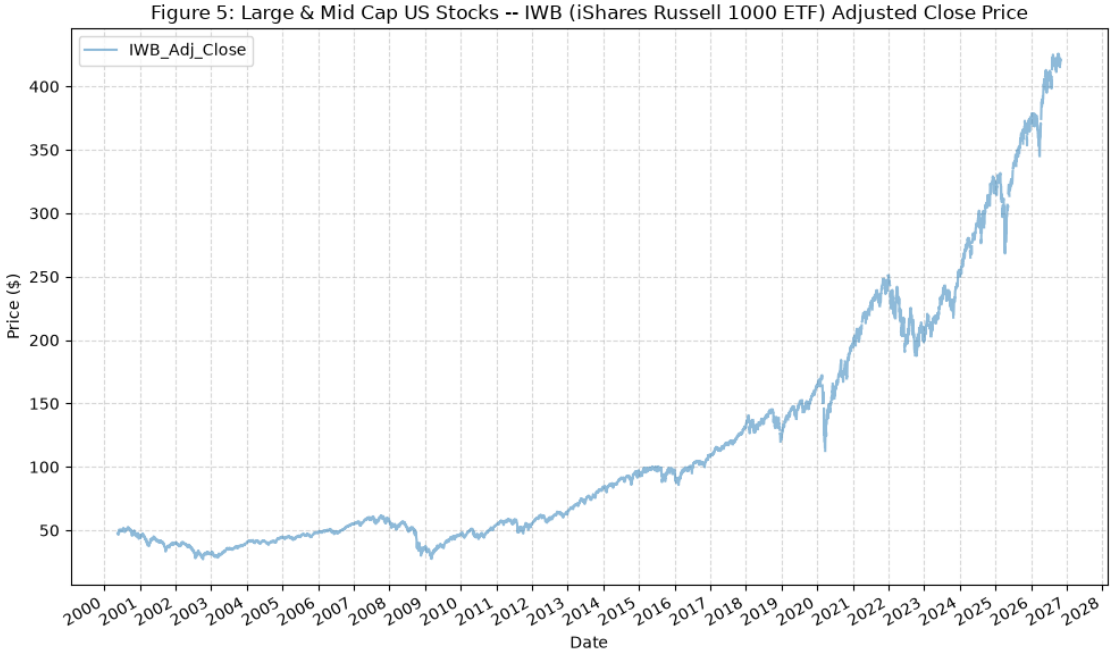
Data Source: Yahoo Finance, Analysis by the Author

Figure 4: US Tech Stocks / Nasdaq 100 -- QQQ (Invesco QQQ Trust, Series 1) Adjusted Close Price



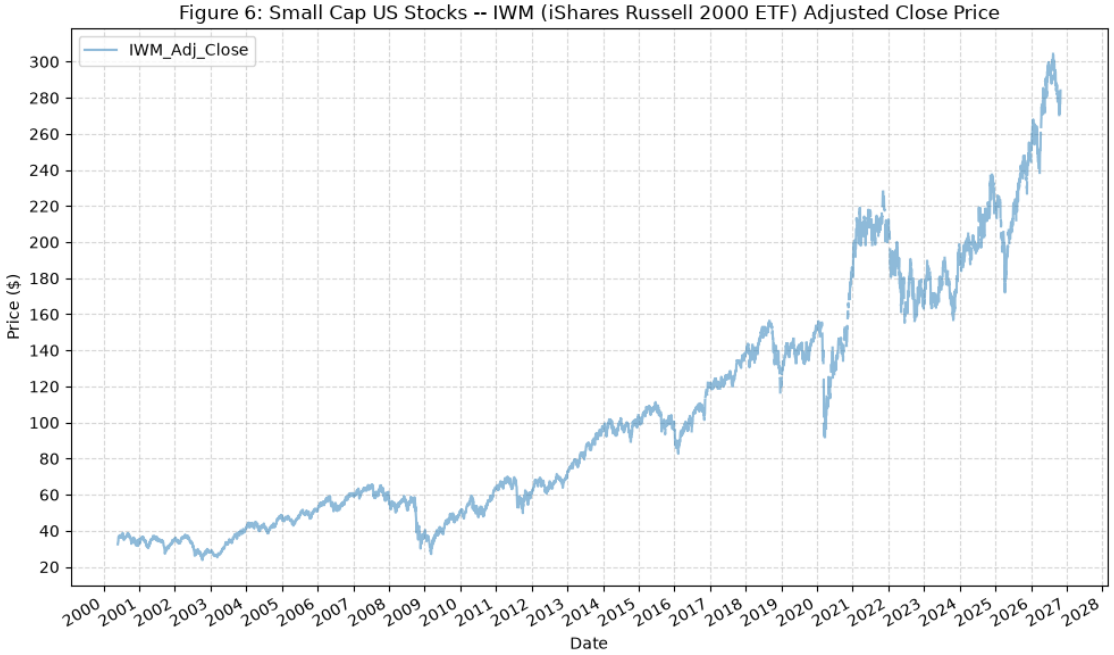
Data Source: Yahoo Finance, Analysis by the Author

Figure 5: Large & Mid Cap US Stocks -- IWB (iShares Russell 1000 ETF) Adjusted Close Price



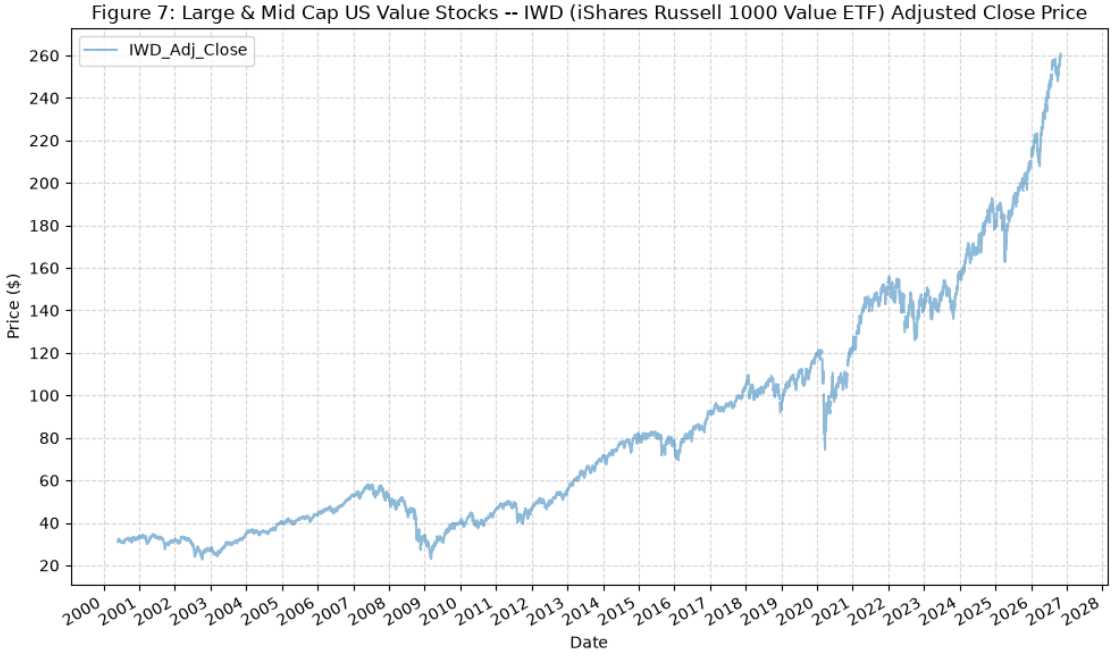
Data Source: Yahoo Finance, Analysis by the Author

Figure 6: Small Cap US Stocks -- IWM (iShares Russell 2000 ETF) Adjusted Close Price



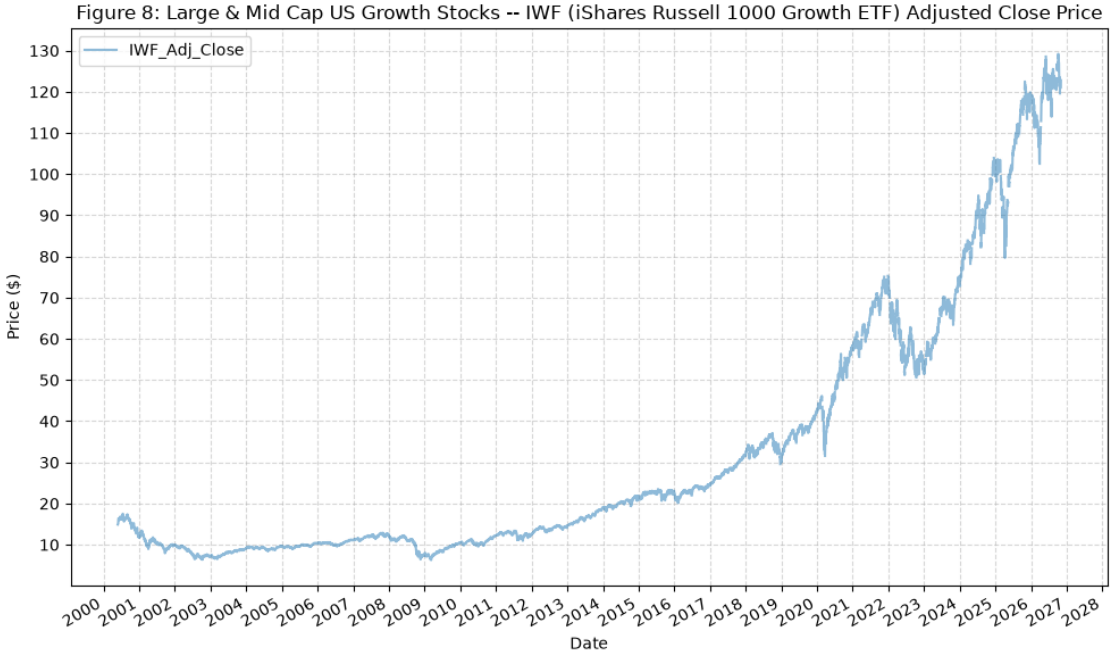
Data Source: Yahoo Finance, Analysis by the Author

Figure 7: Large & Mid Cap US Value Stocks -- IWD (iShares Russell 1000 Value ETF) Adjusted Close Price



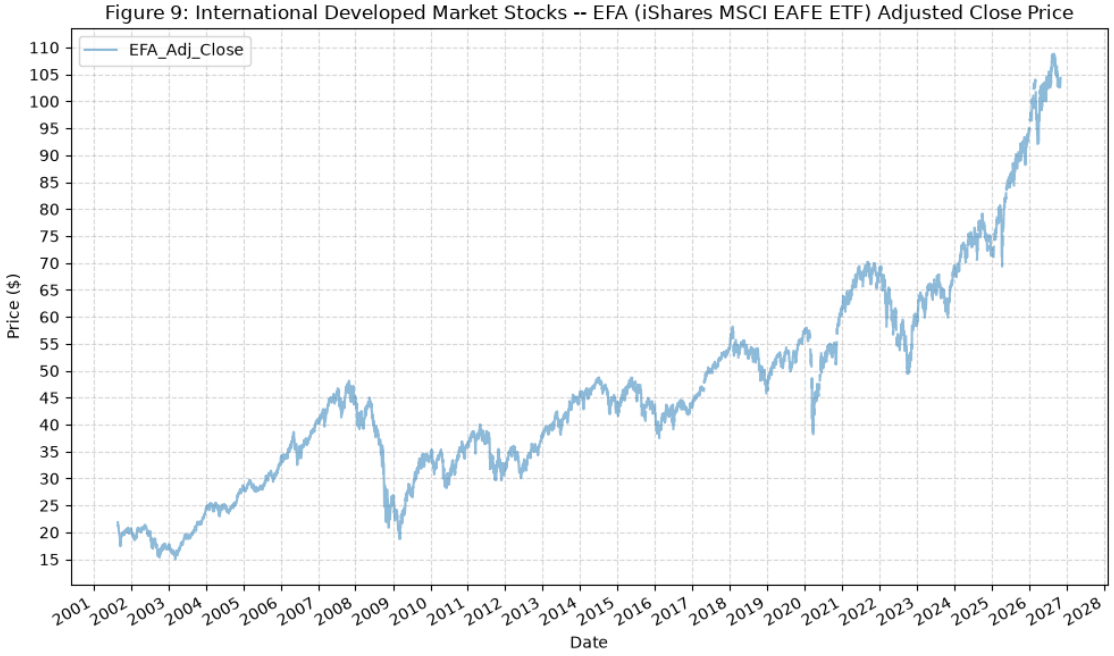
Data Source: Yahoo Finance, Analysis by the Author

Figure 8: Large & Mid Cap US Growth Stocks -- IWF (iShares Russell 1000 Growth ETF) Adjusted Close Price



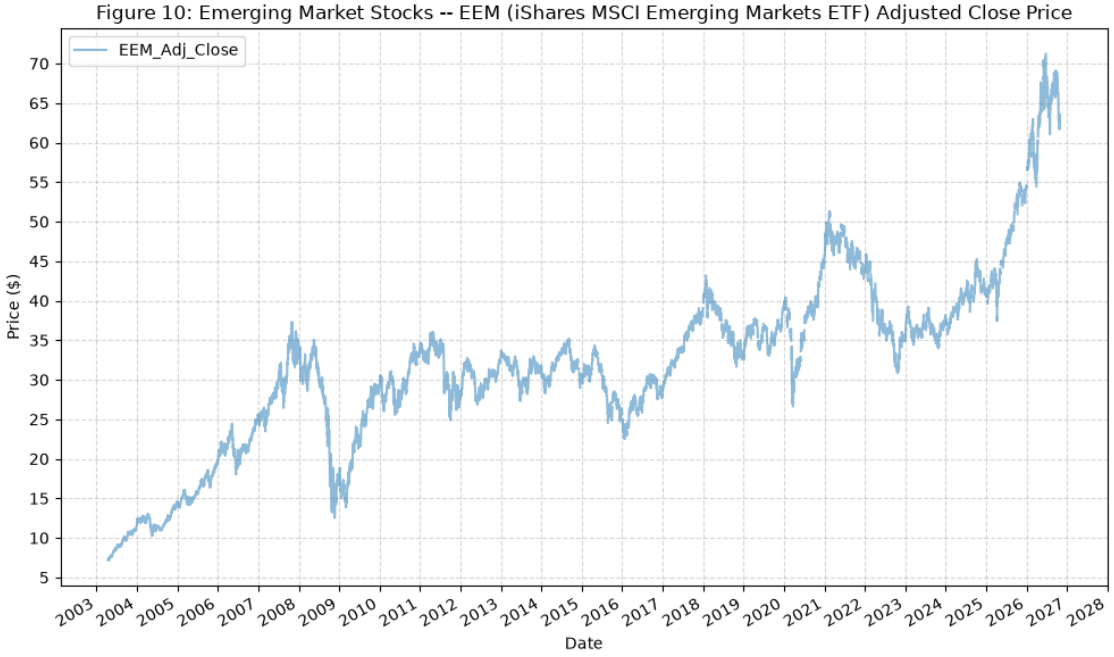
Data Source: Yahoo Finance, Analysis by the Author

Figure 9: International Developed Market Stocks -- EFA (iShares MSCI EAFE ETF) Adjusted Close Price



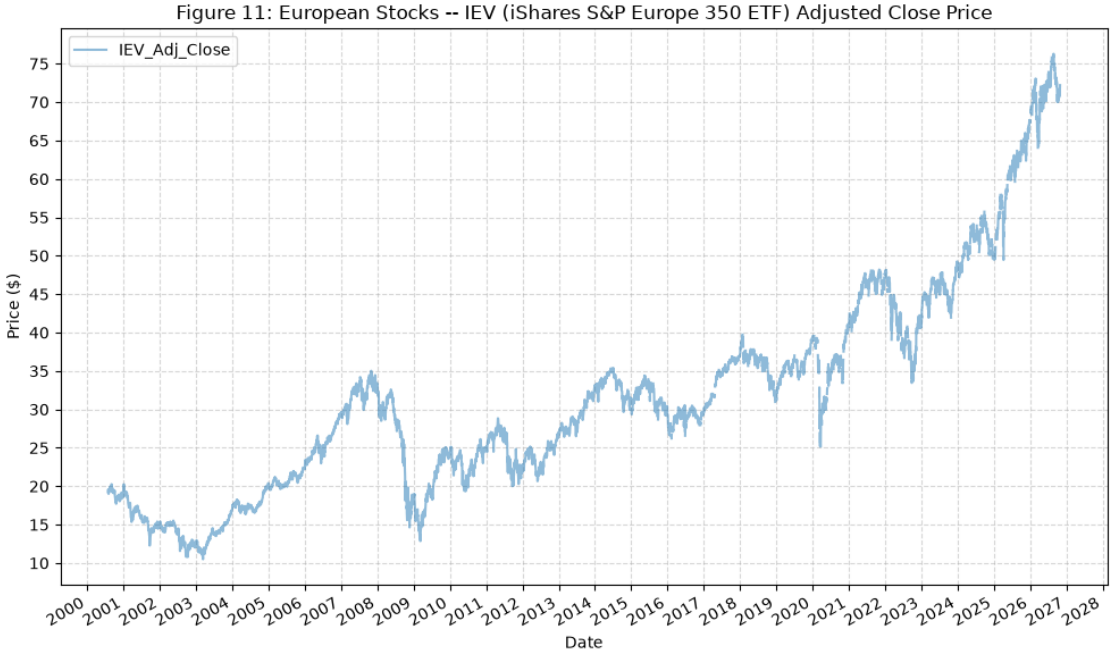
Data Source: Yahoo Finance, Analysis by the Author

Figure 10: Emerging Market Stocks -- EEM (iShares MSCI Emerging Markets ETF) Adjusted Close Price



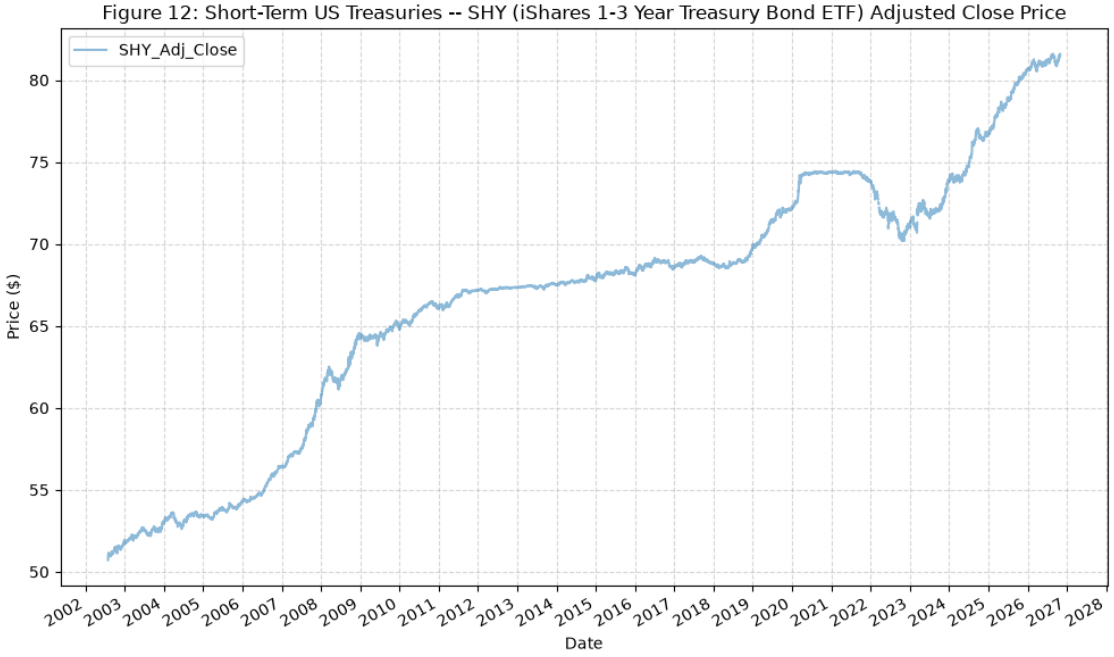
Data Source: Yahoo Finance, Analysis by the Author

Figure 11: European Stocks -- IEV (iShares S&P Europe 350 ETF) Adjusted Close Price



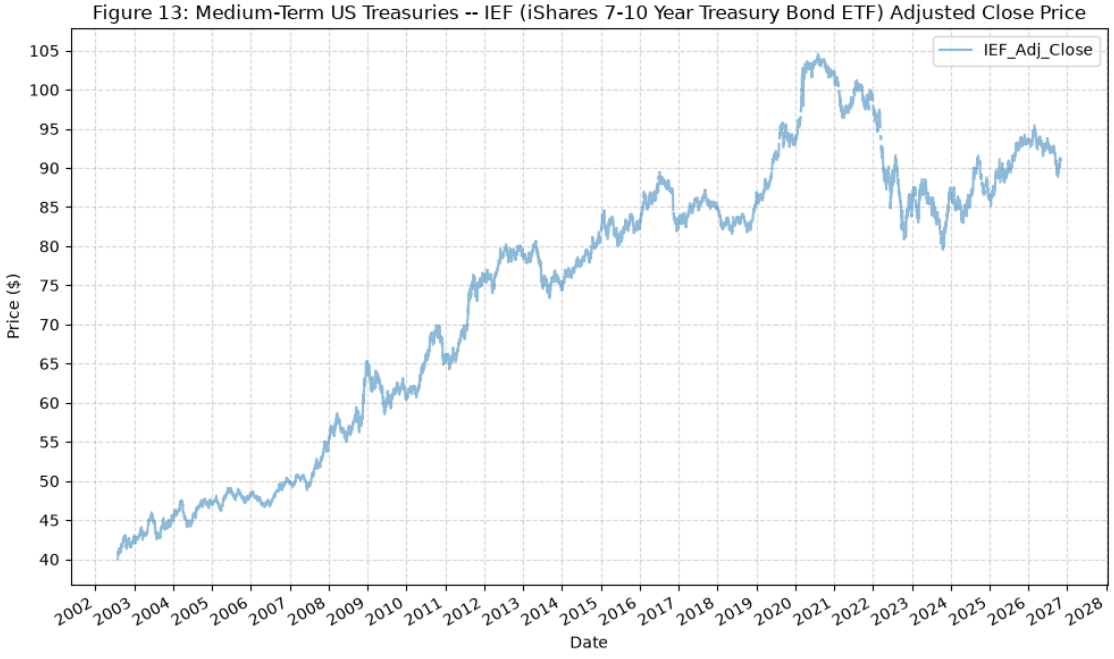
Data Source: Yahoo Finance, Analysis by the Author

Figure 12: Short-Term US Treasuries -- SHY (iShares 1-3 Year Treasury Bond ETF) Adjusted Close Price



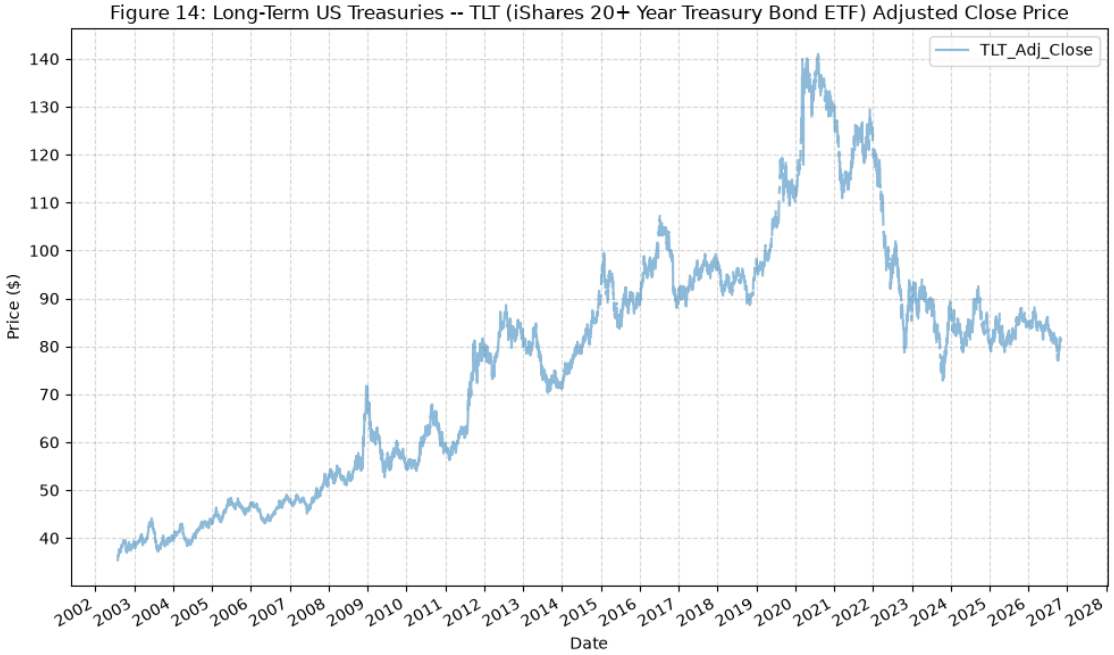
Data Source: Yahoo Finance, Analysis by the Author

Figure 13: Medium-Term US Treasuries -- IEF (iShares 7-10 Year Treasury Bond ETF) Adjusted Close Price



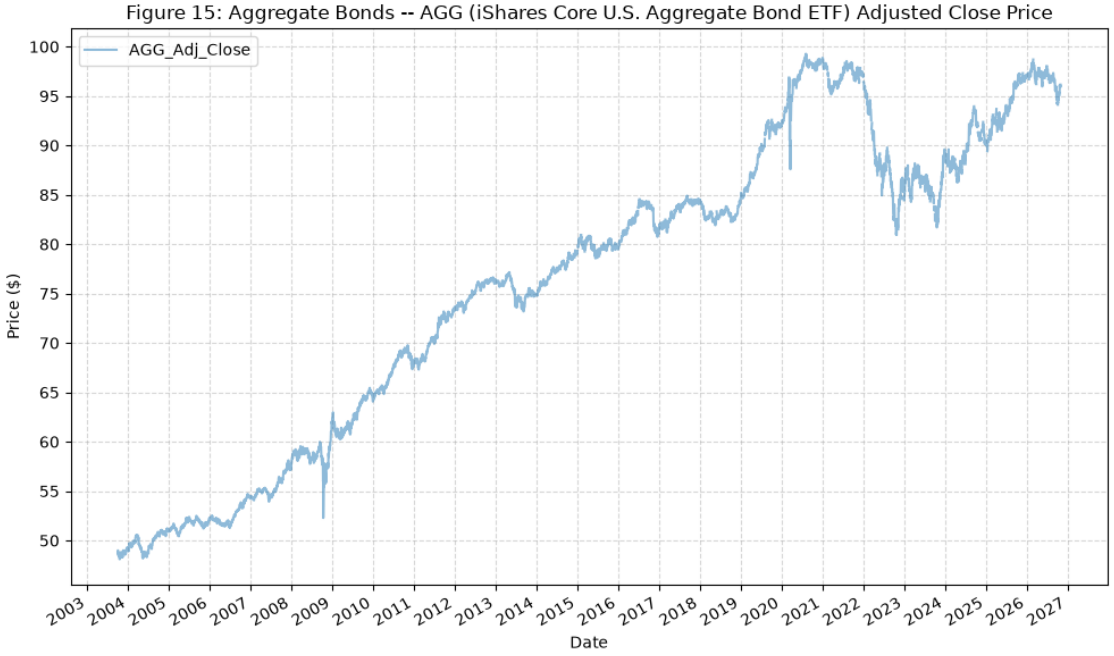
Data Source: Yahoo Finance, Analysis by the Author

Figure 14: Long-Term US Treasuries -- TLT (iShares 20+ Year Treasury Bond ETF) Adjusted Close Price



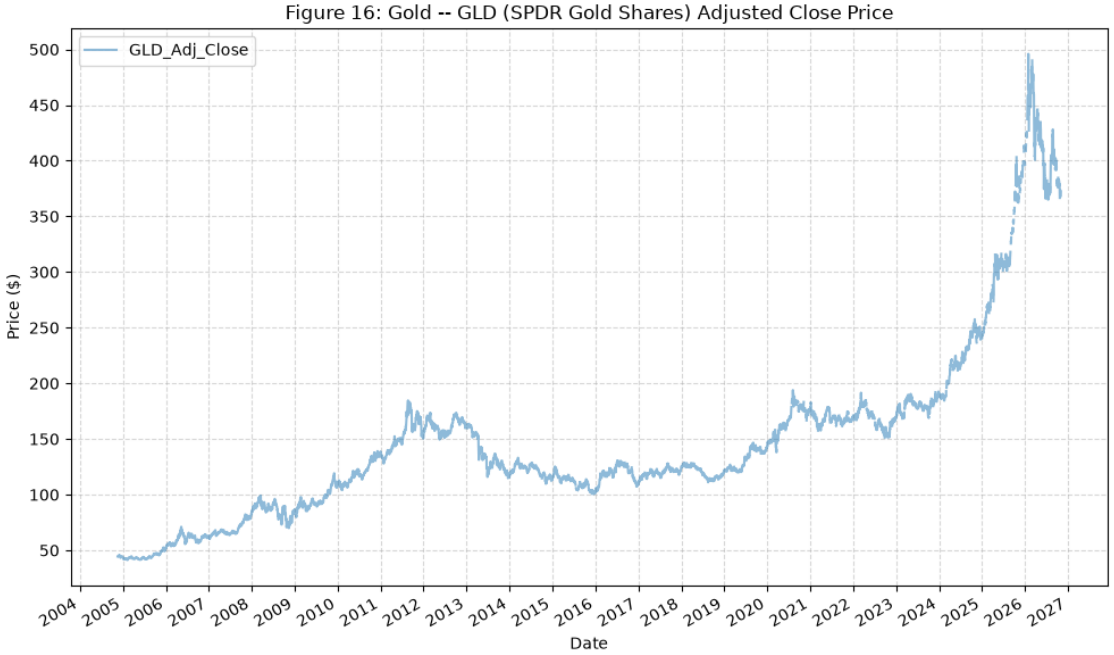
Data Source: Yahoo Finance, Analysis by the Author

Figure 15: Aggregate Bonds -- AGG (iShares Core U.S. Aggregate Bond ETF) Adjusted Close Price



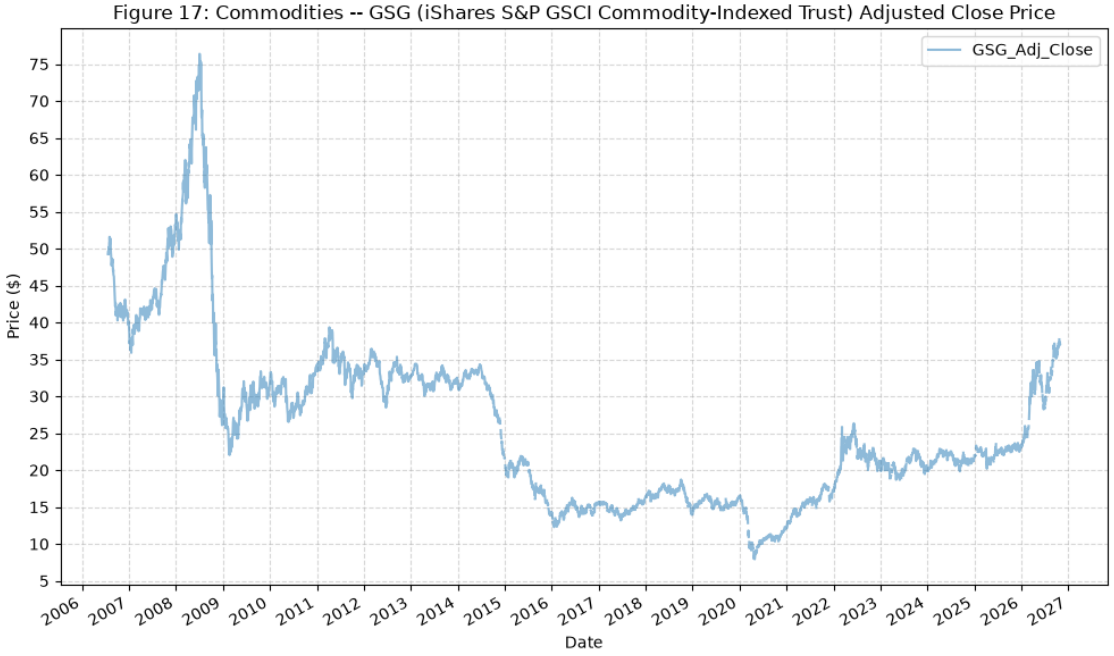
Data Source: Yahoo Finance, Analysis by the Author

Figure 16: Gold -- GLD (SPDR Gold Shares) Adjusted Close Price



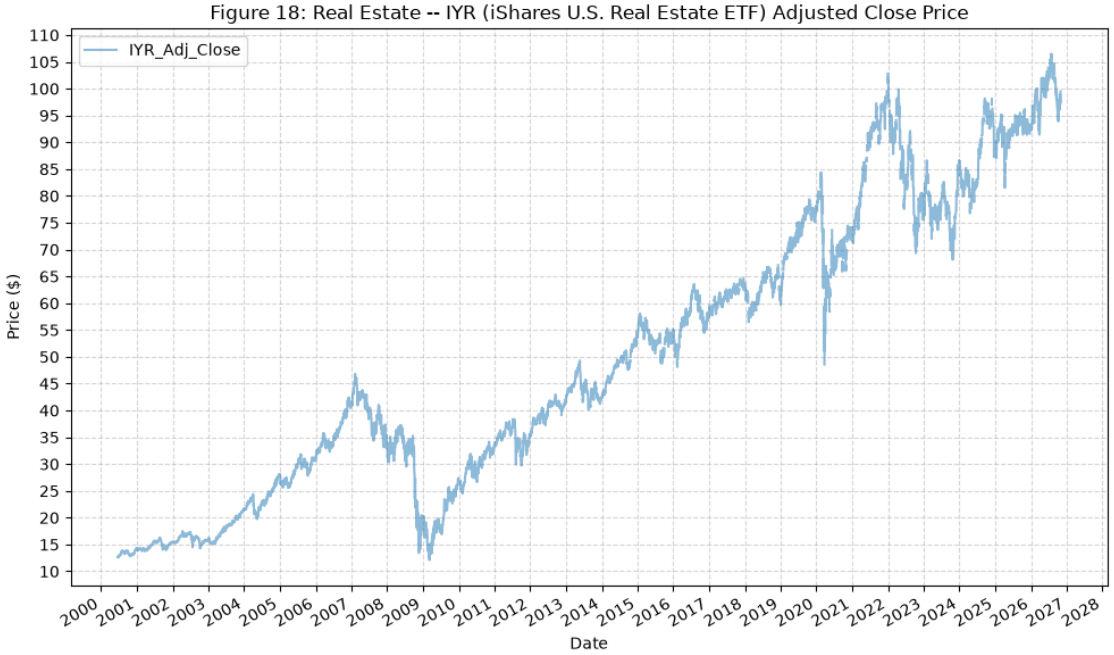
Data Source: Yahoo Finance, Analysis by the Author

Figure 17: Commodities -- GSG (iShares S&P GSCI Commodity-Indexed Trust) Adjusted Close Price



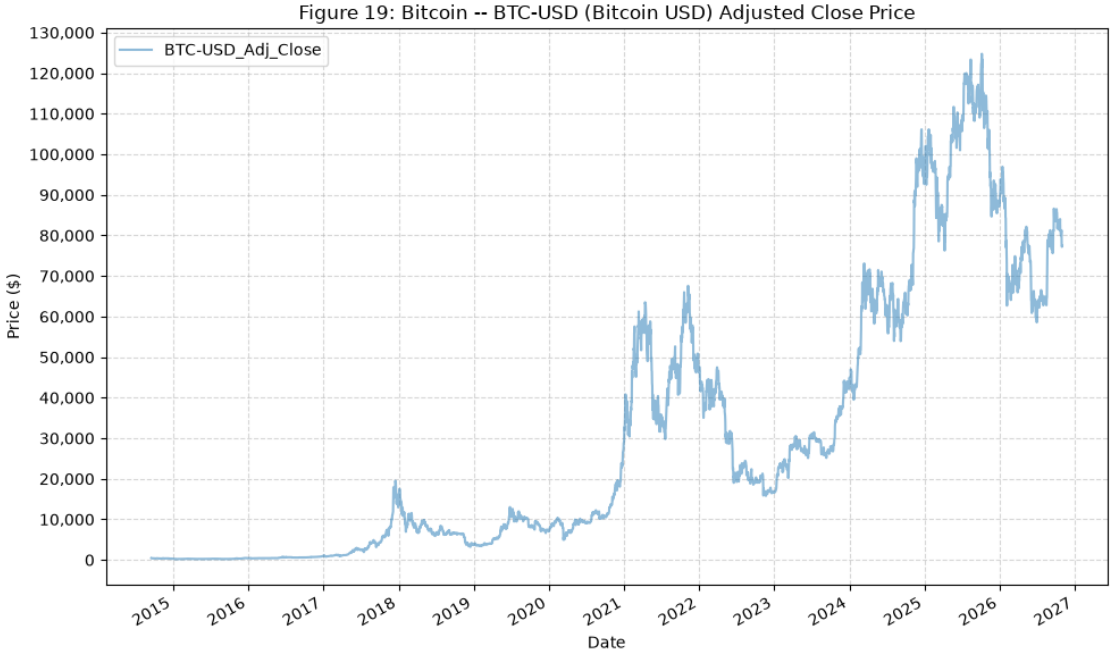
Data Source: Yahoo Finance, Analysis by the Author

Figure 18: Real Estate -- IYR (iShares U.S. Real Estate ETF) Adjusted Close Price



Data Source: Yahoo Finance, Analysis by the Author

Figure 19: Bitcoin -- BTC-USD (Bitcoin USD) Adjusted Close Price



Data Source: Yahoo Finance, Analysis by the Author

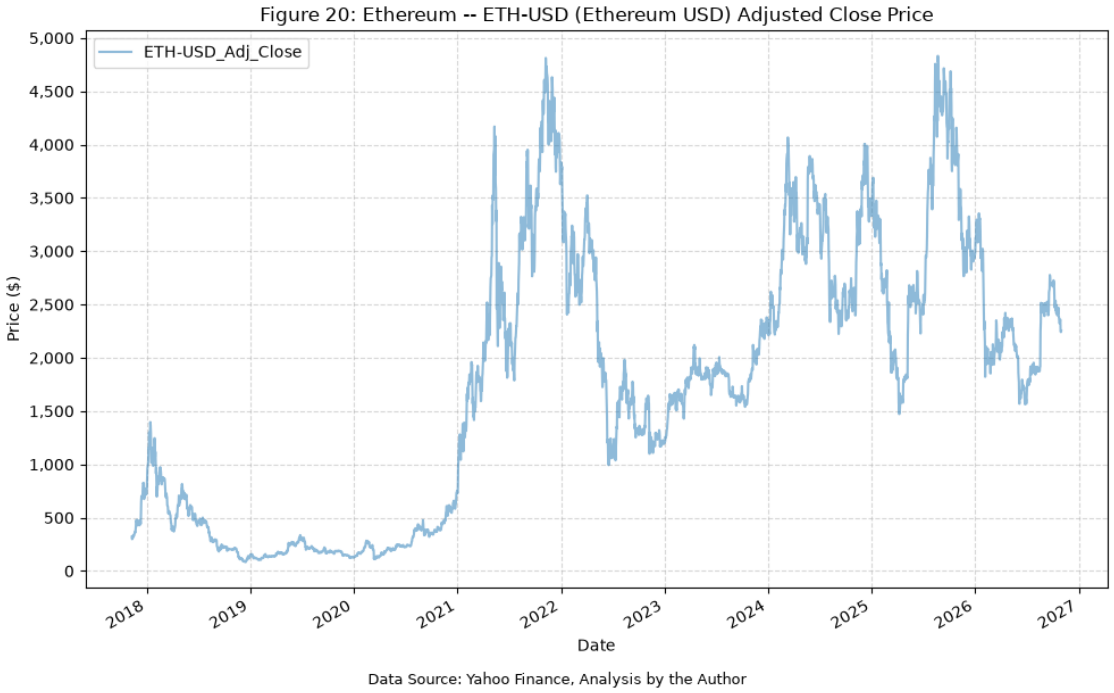


Figure 20: Ethereum -- ETH-USD (Ethereum USD) Adjusted Close Price

Data Source: Yahoo Finance, Analysis by the Author

0.5 Calculate Correlations

Next, we'll calculate the correlation matrix of the daily returns for all of the assets and plot it as a heatmap. We'll do this first without the BTC and ETH data (due to the limited history), and then with the BTC and ETH data.

[9]:
```
# Drop the adjusted close price columns, leaving only the daily return columns
daily_return_columns = [f"{ticker}_Daily_Return" for ticker in tickers_dict.
 ↪keys()]
fund_data_daily_returns_all = fund_data[daily_return_columns]

# Drop the BTC and ETH daily return columns due to the limited history
fund_data_daily_returns_no_crypto = fund_data_daily_returns_all.
 ↪drop(columns=["BTC-USD_Daily_Return", "ETH-USD_Daily_Return"])

# Print the shape of the fund_data_daily_returns_no_crypto DataFrame to confirm␣
 ↪that the rows with missing data have been dropped
print(f"Shape of fund_data_daily_returns_no_crypto:␣
 ↪{fund_data_daily_returns_no_crypto.shape}")
print(f"Rows to drop due to missing data: {fund_data_daily_returns_no_crypto.
 ↪dropna().shape}")

# Drop the NaN values
fund_data_daily_returns_no_crypto = fund_data_daily_returns_no_crypto.dropna()
```

```
# Calculate the correlation matrix of the daily returns
correlation_matrix_no_crypto = fund_data_daily_returns_no_crypto.corr()

display(Markdown(f"Table {next(table_number)}: Correlation Matrix of Daily␣
 ↪Returns (Excluding BTC and ETH)"))
display(correlation_matrix_no_crypto)
```

```
Shape of fund_data_daily_returns_no_crypto: (8344, 18)
Rows to drop due to missing data: (5108, 18)
```

Table 2: Correlation Matrix of Daily Returns (Excluding BTC and ETH)

	IVV_Daily_Return	IJH_Daily_Return	IJR_Daily_Return	\
IVV_Daily_Return	1.00	0.93	0.88	
IJH_Daily_Return	0.93	1.00	0.96	
IJR_Daily_Return	0.88	0.96	1.00	
QQQ_Daily_Return	0.92	0.83	0.78	
IWB_Daily_Return	1.00	0.94	0.89	
IWM_Daily_Return	0.90	0.96	0.98	
IWD_Daily_Return	0.96	0.94	0.90	
IWF_Daily_Return	0.96	0.87	0.82	
EFA_Daily_Return	0.88	0.85	0.80	
EEM_Daily_Return	0.82	0.78	0.73	
IEV_Daily_Return	0.86	0.83	0.78	
SHY_Daily_Return	-0.23	-0.20	-0.19	
IEF_Daily_Return	-0.30	-0.28	-0.27	
TLT_Daily_Return	-0.31	-0.30	-0.29	
AGG_Daily_Return	-0.01	-0.00	-0.03	
GLD_Daily_Return	0.07	0.08	0.05	
GSG_Daily_Return	0.37	0.37	0.34	
IYR_Daily_Return	0.76	0.78	0.76	

	QQQ_Daily_Return	IWB_Daily_Return	IWM_Daily_Return	\
IVV_Daily_Return	0.92	1.00	0.90	
IJH_Daily_Return	0.83	0.94	0.96	
IJR_Daily_Return	0.78	0.89	0.98	
QQQ_Daily_Return	1.00	0.93	0.82	
IWB_Daily_Return	0.93	1.00	0.91	
IWM_Daily_Return	0.82	0.91	1.00	
IWD_Daily_Return	0.81	0.96	0.90	
IWF_Daily_Return	0.97	0.96	0.85	
EFA_Daily_Return	0.78	0.88	0.81	
EEM_Daily_Return	0.76	0.82	0.76	
IEV_Daily_Return	0.76	0.86	0.80	
SHY_Daily_Return	-0.19	-0.22	-0.19	
IEF_Daily_Return	-0.24	-0.29	-0.27	
TLT_Daily_Return	-0.25	-0.30	-0.28	
AGG_Daily_Return	0.00	-0.01	-0.02	

GLD_Daily_Return	0.06	0.07	0.07
GSG_Daily_Return	0.29	0.37	0.35
IYR_Daily_Return	0.63	0.76	0.76

	IWD_Daily_Return	IWF_Daily_Return	EFA_Daily_Return	\
IVV_Daily_Return	0.96	0.96	0.88	
IJH_Daily_Return	0.94	0.87	0.85	
IJR_Daily_Return	0.90	0.82	0.80	
QQQ_Daily_Return	0.81	0.97	0.78	
IWB_Daily_Return	0.96	0.96	0.88	
IWM_Daily_Return	0.90	0.85	0.81	
IWD_Daily_Return	1.00	0.86	0.88	
IWF_Daily_Return	0.86	1.00	0.82	
EFA_Daily_Return	0.88	0.82	1.00	
EEM_Daily_Return	0.80	0.78	0.87	
IEV_Daily_Return	0.87	0.80	0.98	
SHY_Daily_Return	-0.24	-0.19	-0.18	
IEF_Daily_Return	-0.31	-0.25	-0.25	
TLT_Daily_Return	-0.33	-0.26	-0.28	
AGG_Daily_Return	-0.01	0.02	0.04	
GLD_Daily_Return	0.06	0.08	0.17	
GSG_Daily_Return	0.39	0.33	0.40	
IYR_Daily_Return	0.80	0.67	0.70	

	EEM_Daily_Return	IEV_Daily_Return	SHY_Daily_Return	\
IVV_Daily_Return	0.82	0.86	-0.23	
IJH_Daily_Return	0.78	0.83	-0.20	
IJR_Daily_Return	0.73	0.78	-0.19	
QQQ_Daily_Return	0.76	0.76	-0.19	
IWB_Daily_Return	0.82	0.86	-0.22	
IWM_Daily_Return	0.76	0.80	-0.19	
IWD_Daily_Return	0.80	0.87	-0.24	
IWF_Daily_Return	0.78	0.80	-0.19	
EFA_Daily_Return	0.87	0.98	-0.18	
EEM_Daily_Return	1.00	0.84	-0.21	
IEV_Daily_Return	0.84	1.00	-0.18	
SHY_Daily_Return	-0.21	-0.18	1.00	
IEF_Daily_Return	-0.26	-0.26	0.76	
TLT_Daily_Return	-0.26	-0.29	0.57	
AGG_Daily_Return	-0.01	0.03	0.60	
GLD_Daily_Return	0.19	0.16	0.22	
GSG_Daily_Return	0.40	0.40	-0.14	
IYR_Daily_Return	0.66	0.69	-0.12	

	IEF_Daily_Return	TLT_Daily_Return	AGG_Daily_Return	\
IVV_Daily_Return	-0.30	-0.31	-0.01	
IJH_Daily_Return	-0.28	-0.30	-0.00	
IJR_Daily_Return	-0.27	-0.29	-0.03	

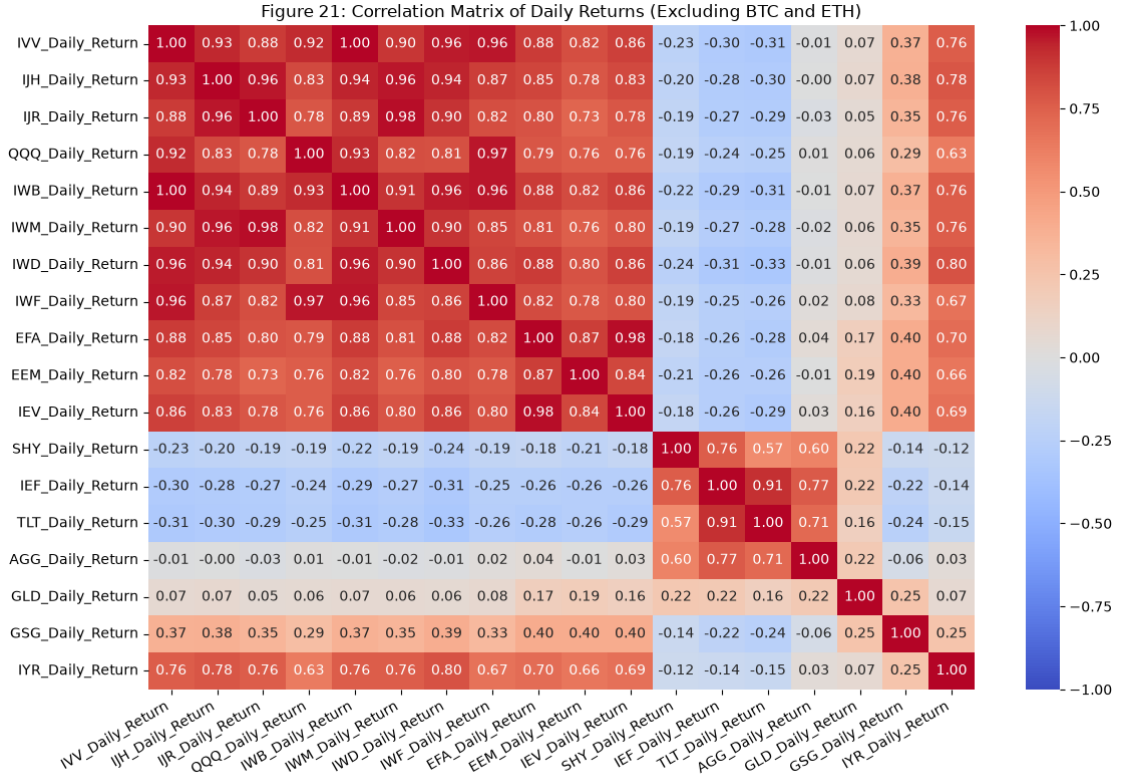
QQQ_Daily_Return	-0.24	-0.25	0.00
IWB_Daily_Return	-0.29	-0.30	-0.01
IWM_Daily_Return	-0.27	-0.28	-0.02
IWD_Daily_Return	-0.31	-0.33	-0.01
IWF_Daily_Return	-0.25	-0.26	0.02
EFA_Daily_Return	-0.25	-0.28	0.04
EEM_Daily_Return	-0.26	-0.26	-0.01
IEV_Daily_Return	-0.26	-0.29	0.03
SHY_Daily_Return	0.76	0.57	0.60
IEF_Daily_Return	1.00	0.91	0.77
TLT_Daily_Return	0.91	1.00	0.71
AGG_Daily_Return	0.77	0.71	1.00
GLD_Daily_Return	0.22	0.16	0.22
GSG_Daily_Return	-0.21	-0.24	-0.06
IYR_Daily_Return	-0.14	-0.15	0.03

	GLD_Daily_Return	GSG_Daily_Return	IYR_Daily_Return
IVV_Daily_Return	0.07	0.37	0.76
IJH_Daily_Return	0.08	0.37	0.78
IJR_Daily_Return	0.05	0.34	0.76
QQQ_Daily_Return	0.06	0.29	0.63
IWB_Daily_Return	0.07	0.37	0.76
IWM_Daily_Return	0.07	0.35	0.76
IWD_Daily_Return	0.06	0.39	0.80
IWF_Daily_Return	0.08	0.33	0.67
EFA_Daily_Return	0.17	0.40	0.70
EEM_Daily_Return	0.19	0.40	0.66
IEV_Daily_Return	0.16	0.40	0.69
SHY_Daily_Return	0.22	-0.14	-0.12
IEF_Daily_Return	0.22	-0.21	-0.14
TLT_Daily_Return	0.16	-0.24	-0.15
AGG_Daily_Return	0.22	-0.06	0.03
GLD_Daily_Return	1.00	0.25	0.07
GSG_Daily_Return	0.25	1.00	0.25
IYR_Daily_Return	0.07	0.25	1.00

And then the heatmap:

[10]:
```
plot_heatmap(
    df=correlation_matrix_no_crypto,
    title=f"Figure {next(figure_number)}: Correlation Matrix of Daily Returns␣
 ↪(Excluding BTC and ETH)",
    caption=f"Data Source: Yahoo Finance, Analysis by the Author",
)
```

Figure 21: Correlation Matrix of Daily Returns (Excluding BTC and ETH)



Data Source: Yahoo Finance, Analysis by the Author

We'll now include the BTC and ETH data and recalculate the correlation matrix and heatmap. Keep in mind that the BTC and ETH data only goes back to 2014 and 2017 respectively, so the correlation matrix will be calculated using a shorter time period than the other assets.

[11]:
```
# Print the shape of the fund_data_daily_returns DataFrame to confirm that the␣
 ↪rows with missing data have been dropped
print(f"Shape of fund_data_daily_returns: {fund_data_daily_returns_all.shape}")
print(f"Rows to drop due to missing data: {fund_data_daily_returns_all.dropna().
 ↪shape}")

# Drop the NaN values
fund_data_daily_returns = fund_data_daily_returns_all.dropna()

# Calculate the correlation matrix of the daily returns
correlation_matrix = fund_data_daily_returns.corr()

display(Markdown(f"Table {next(table_number)}: Correlation Matrix of Daily␣
 ↪Returns (Including BTC and ETH)"))
display(correlation_matrix)
```

```
Shape of fund_data_daily_returns: (8344, 20)
Rows to drop due to missing data: (2261, 20)
```

Table 3: Correlation Matrix of Daily Returns (Including BTC and ETH)

	IVV_Daily_Return	IJH_Daily_Return	IJR_Daily_Return	\
IVV_Daily_Return	1.00	0.90	0.84	
IJH_Daily_Return	0.90	1.00	0.96	
IJR_Daily_Return	0.84	0.96	1.00	
QQQ_Daily_Return	0.93	0.77	0.71	
IWB_Daily_Return	1.00	0.91	0.85	
IWM_Daily_Return	0.86	0.96	0.98	
IWD_Daily_Return	0.92	0.94	0.90	
IWF_Daily_Return	0.96	0.80	0.73	
EFA_Daily_Return	0.85	0.83	0.78	
EEM_Daily_Return	0.75	0.71	0.66	
IEV_Daily_Return	0.82	0.81	0.76	
SHY_Daily_Return	-0.04	-0.04	-0.03	
IEF_Daily_Return	-0.11	-0.10	-0.09	
TLT_Daily_Return	-0.14	-0.13	-0.12	
AGG_Daily_Return	0.14	0.16	0.16	
GLD_Daily_Return	0.12	0.11	0.10	
GSG_Daily_Return	0.28	0.29	0.28	
IYR_Daily_Return	0.72	0.77	0.73	
BTC-USD_Daily_Return	0.30	0.29	0.28	
ETH-USD_Daily_Return	0.33	0.30	0.30	

	QQQ_Daily_Return	IWB_Daily_Return	IWM_Daily_Return	\
IVV_Daily_Return	0.93	1.00	0.86	
IJH_Daily_Return	0.77	0.91	0.96	
IJR_Daily_Return	0.71	0.85	0.98	
QQQ_Daily_Return	1.00	0.93	0.77	
IWB_Daily_Return	0.93	1.00	0.88	
IWM_Daily_Return	0.77	0.88	1.00	
IWD_Daily_Return	0.75	0.92	0.89	
IWF_Daily_Return	0.98	0.96	0.79	
EFA_Daily_Return	0.76	0.85	0.80	
EEM_Daily_Return	0.74	0.75	0.70	
IEV_Daily_Return	0.71	0.82	0.77	
SHY_Daily_Return	-0.02	-0.04	-0.01	
IEF_Daily_Return	-0.07	-0.10	-0.07	
TLT_Daily_Return	-0.09	-0.13	-0.10	
AGG_Daily_Return	0.14	0.15	0.17	
GLD_Daily_Return	0.14	0.13	0.13	
GSG_Daily_Return	0.21	0.28	0.27	
IYR_Daily_Return	0.56	0.73	0.72	
BTC-USD_Daily_Return	0.31	0.31	0.31	
ETH-USD_Daily_Return	0.34	0.33	0.33	

	IWD_Daily_Return	IWF_Daily_Return	EFA_Daily_Return	\
IVV_Daily_Return	0.92	0.96	0.85	

IJH_Daily_Return	0.94	0.80	0.83
IJR_Daily_Return	0.90	0.73	0.78
QQQ_Daily_Return	0.75	0.98	0.76
IWB_Daily_Return	0.92	0.96	0.85
IWM_Daily_Return	0.89	0.79	0.80
IWD_Daily_Return	1.00	0.78	0.85
IWF_Daily_Return	0.78	1.00	0.77
EFA_Daily_Return	0.85	0.77	1.00
EEM_Daily_Return	0.69	0.73	0.82
IEV_Daily_Return	0.83	0.74	0.98
SHY_Daily_Return	-0.05	-0.02	0.03
IEF_Daily_Return	-0.13	-0.07	-0.03
TLT_Daily_Return	-0.17	-0.09	-0.08
AGG_Daily_Return	0.13	0.15	0.22
GLD_Daily_Return	0.11	0.13	0.25
GSG_Daily_Return	0.32	0.22	0.26
IYR_Daily_Return	0.80	0.61	0.68
BTC-USD_Daily_Return	0.26	0.31	0.29
ETH-USD_Daily_Return	0.28	0.34	0.32

	EEM_Daily_Return	IEV_Daily_Return	SHY_Daily_Return	\
IVV_Daily_Return	0.75	0.82	-0.04	
IJH_Daily_Return	0.71	0.81	-0.04	
IJR_Daily_Return	0.66	0.76	-0.03	
QQQ_Daily_Return	0.74	0.71	-0.02	
IWB_Daily_Return	0.75	0.82	-0.04	
IWM_Daily_Return	0.70	0.77	-0.01	
IWD_Daily_Return	0.69	0.83	-0.05	
IWF_Daily_Return	0.73	0.74	-0.02	
EFA_Daily_Return	0.82	0.98	0.03	
EEM_Daily_Return	1.00	0.78	0.01	
IEV_Daily_Return	0.78	1.00	0.02	
SHY_Daily_Return	0.01	0.02	1.00	
IEF_Daily_Return	-0.05	-0.04	0.82	
TLT_Daily_Return	-0.08	-0.08	0.60	
AGG_Daily_Return	0.16	0.21	0.74	
GLD_Daily_Return	0.27	0.24	0.32	
GSG_Daily_Return	0.25	0.25	-0.12	
IYR_Daily_Return	0.51	0.67	0.14	
BTC-USD_Daily_Return	0.27	0.29	0.03	
ETH-USD_Daily_Return	0.30	0.31	0.03	

	IEF_Daily_Return	TLT_Daily_Return	AGG_Daily_Return	\
IVV_Daily_Return	-0.11	-0.14	0.14	
IJH_Daily_Return	-0.10	-0.13	0.16	
IJR_Daily_Return	-0.09	-0.12	0.16	
QQQ_Daily_Return	-0.07	-0.09	0.14	
IWB_Daily_Return	-0.10	-0.13	0.15	

IWM_Daily_Return	-0.07	-0.10	0.17
IWD_Daily_Return	-0.13	-0.17	0.13
IWF_Daily_Return	-0.07	-0.09	0.15
EFA_Daily_Return	-0.03	-0.08	0.22
EEM_Daily_Return	-0.05	-0.08	0.16
IEV_Daily_Return	-0.04	-0.08	0.21
SHY_Daily_Return	0.82	0.60	0.74
IEF_Daily_Return	1.00	0.91	0.89
TLT_Daily_Return	0.91	1.00	0.84
AGG_Daily_Return	0.89	0.84	1.00
GLD_Daily_Return	0.30	0.22	0.30
GSG_Daily_Return	-0.16	-0.18	-0.04
IYR_Daily_Return	0.10	0.05	0.30
BTC-USD_Daily_Return	-0.00	-0.01	0.12
ETH-USD_Daily_Return	-0.00	-0.01	0.12

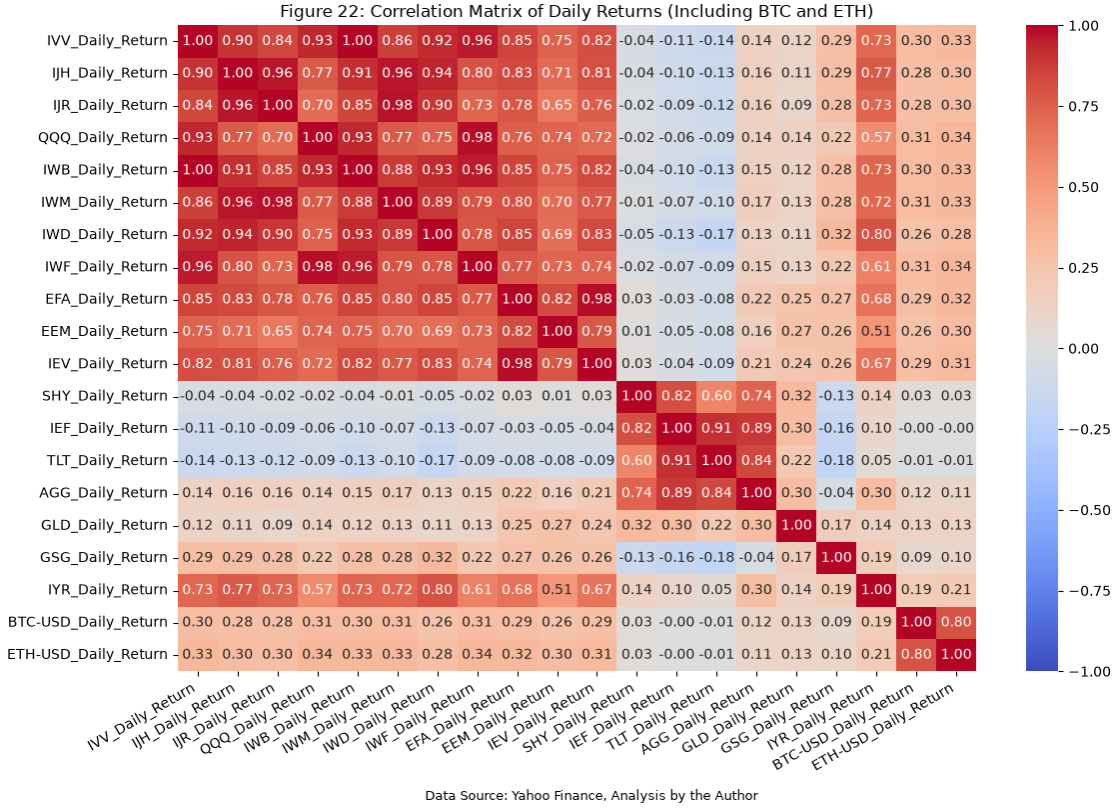
	GLD_Daily_Return	GSG_Daily_Return	IYR_Daily_Return	\
IVV_Daily_Return	0.12	0.28	0.72	
IJH_Daily_Return	0.11	0.29	0.77	
IJR_Daily_Return	0.10	0.28	0.73	
QQQ_Daily_Return	0.14	0.21	0.56	
IWB_Daily_Return	0.13	0.28	0.73	
IWM_Daily_Return	0.13	0.27	0.72	
IWD_Daily_Return	0.11	0.32	0.80	
IWF_Daily_Return	0.13	0.22	0.61	
EFA_Daily_Return	0.25	0.26	0.68	
EEM_Daily_Return	0.27	0.25	0.51	
IEV_Daily_Return	0.24	0.25	0.67	
SHY_Daily_Return	0.32	-0.12	0.14	
IEF_Daily_Return	0.30	-0.16	0.10	
TLT_Daily_Return	0.22	-0.18	0.05	
AGG_Daily_Return	0.30	-0.04	0.30	
GLD_Daily_Return	1.00	0.17	0.14	
GSG_Daily_Return	0.17	1.00	0.19	
IYR_Daily_Return	0.14	0.19	1.00	
BTC-USD_Daily_Return	0.13	0.08	0.19	
ETH-USD_Daily_Return	0.13	0.10	0.21	

	BTC-USD_Daily_Return	ETH-USD_Daily_Return
IVV_Daily_Return	0.30	0.33
IJH_Daily_Return	0.29	0.30
IJR_Daily_Return	0.28	0.30
QQQ_Daily_Return	0.31	0.34
IWB_Daily_Return	0.31	0.33
IWM_Daily_Return	0.31	0.33
IWD_Daily_Return	0.26	0.28
IWF_Daily_Return	0.31	0.34
EFA_Daily_Return	0.29	0.32

EEM_Daily_Return	0.27	0.30
IEV_Daily_Return	0.29	0.31
SHY_Daily_Return	0.03	0.03
IEF_Daily_Return	-0.00	-0.00
TLT_Daily_Return	-0.01	-0.01
AGG_Daily_Return	0.12	0.12
GLD_Daily_Return	0.13	0.13
GSG_Daily_Return	0.08	0.10
IYR_Daily_Return	0.19	0.21
BTC-USD_Daily_Return	1.00	0.80
ETH-USD_Daily_Return	0.80	1.00

And then the heatmap:

[12]:
```
plot_heatmap(
    df=correlation_matrix,
    title=f"Figure {next(figure_number)}: Correlation Matrix of Daily Returns␣
 ↪(Including BTC and ETH)",
    caption=f"Data Source: Yahoo Finance, Analysis by the Author",
)
```



These are interesting results, but expected. The stock funds tend to have low correlations with

the bond funds, the commodities don't really correlate with anything, etc. But we know that the correlations between these asset classes have changed over time, so let's take a look at how the correlations have evolved over time by calculating rolling correlations.

0.6 Calculate Rolling Correlations

Next, we will calculate the rolling correlations for several different periods:

- 3 days
- 1 week
- 2 weeks
- 1 month
- 3 month
- 6 month
- 1 year
- 5 years
- 10 years

[13]:
```
# Define rolling windows in trading days
rolling_windows = {
    "3d": 3,  # 3 days (~3 trading days)
    "1w": 5,  # 1 week (~5 trading days)
    "2w": 10,  # 2 weeks (~10 trading days)
    "1m": 21,  # 1 month (~21 trading days)
    "3m": 63,  # 3 months (~63 trading days)
    "6m": 126,  # 6 months (~126 trading days)
    "1y": 252,  # 1 year (~252 trading days)
    "5y": 1260,  # 5 years (~1260 trading days)
    "10y": 2520,  # 10 years (~2520 trading days)
}
```

We'll pair down the list of funds to only include the high correlation US equity funds that make up the S&P 1500 index (IVV, IJH, IJR).

[14]:
```
# Create temp list for tickers
tickers = ["IVV", "IJH", "IJR"]
pairs = list(combinations(tickers, 2))

# Create empty dictionary to store rolling correlation results
rolling_corr_results = {}

# Create empty DataFrame to store rolling correlation results
rolling_corr_results_df = pd.DataFrame()

# Calculate rolling correlations for each pair of tickers and each rolling␣
 ↪window
for ticker1, ticker2 in pairs:
    try:
        temp_df = fund_data_daily_returns_no_crypto[
```

```
            [f"{ticker1}_Daily_Return", f"{ticker2}_Daily_Return"]
        ].dropna()
    except Exception as e:
        print(f"Error creating temp_df for {ticker1} and {ticker2}: {e}")
        continue

    for window_name, window_size in rolling_windows.items():
        try:
            rolling_corr = (
                temp_df[f"{ticker1}_Daily_Return"]
                .rolling(window=window_size)
                .corr(temp_df[f"{ticker2}_Daily_Return"])
            )

            # print(tickers_dict[f"{ticker1}"])
            # print(tickers_dict[f"{ticker2}"])

            rolling_corr_results[
                f"{ticker1}_{ticker2}_{window_name}"
            ] = rolling_corr
            rolling_corr_results_df = pd.concat(
                [
                    rolling_corr_results_df,
                    rolling_corr.
 ↪to_frame(name=f"{ticker1}_{ticker2}_{window_name}"),
                ],
                axis=1,
            )

        except Exception as e:
            print(
                f"Error calculating rolling correlation for {ticker1} and␣
 ↪{ticker2} with window {window_name}: {e}"
            )

display(Markdown(f"Table {next(table_number)}: Rolling Correlations"))
display(rolling_corr_results_df)
```

Table 4: Rolling Correlations

	IVV_IJH_3d	IVV_IJH_1w	IVV_IJH_2w	IVV_IJH_1m	IVV_IJH_3m	\
Date						
2006-07-24	NaN	NaN	NaN	NaN	NaN	
2006-07-25	NaN	NaN	NaN	NaN	NaN	
2006-07-26	0.97	NaN	NaN	NaN	NaN	
2006-07-27	0.99	NaN	NaN	NaN	NaN	
2006-07-28	1.00	0.96	NaN	NaN	NaN	
…	…	…	…	…	…	

```
2026-11-04       -0.23        0.58        0.67        0.64        0.70
2026-11-05        0.99        0.93        0.90        0.88        0.80
2026-11-06        0.98        0.93        0.90        0.89        0.80
2026-11-09        1.00        0.96        0.92        0.90        0.83
2026-11-10        1.00        0.96        0.92        0.91        0.83

            IVV_IJH_6m  IVV_IJH_1y  IVV_IJH_5y  IVV_IJH_10y  IVV_IJR_3d  …  \
Date                                                                       …
2006-07-24         NaN         NaN         NaN          NaN         NaN  …
2006-07-25         NaN         NaN         NaN          NaN         NaN  …
2006-07-26         NaN         NaN         NaN          NaN        0.96  …
2006-07-27         NaN         NaN         NaN          NaN        0.93  …
2006-07-28         NaN         NaN         NaN          NaN        0.99  …
…                  …           …           …            …          …     …
2026-11-04        0.75        0.80        0.88         0.90        0.40  …
2026-11-05        0.79        0.81        0.88         0.90        0.99  …
2026-11-06        0.79        0.81        0.88         0.90        1.00  …
2026-11-09        0.81        0.82        0.88         0.90        0.99  …
2026-11-10        0.81        0.82        0.88         0.90        0.99  …

            IVV_IJR_10y  IJH_IJR_3d  IJH_IJR_1w  IJH_IJR_2w  IJH_IJR_1m  \
Date
2006-07-24          NaN         NaN         NaN         NaN         NaN
2006-07-25          NaN         NaN         NaN         NaN         NaN
2006-07-26          NaN        1.00         NaN         NaN         NaN
2006-07-27          NaN        0.97         NaN         NaN         NaN
2006-07-28          NaN        0.98        0.99         NaN         NaN
…                   …          …           …           …           …
2026-11-04         0.83        0.80        0.92        0.76        0.87
2026-11-05         0.83        0.97        0.97        0.97        0.93
2026-11-06         0.83        0.97        0.97        0.97        0.93
2026-11-09         0.83        1.00        0.98        0.98        0.95
2026-11-10         0.83        0.99        0.98        0.98        0.95

            IJH_IJR_3m  IJH_IJR_6m  IJH_IJR_1y  IJH_IJR_5y  IJH_IJR_10y
Date
2006-07-24         NaN         NaN         NaN         NaN          NaN
2006-07-25         NaN         NaN         NaN         NaN          NaN
2006-07-26         NaN         NaN         NaN         NaN          NaN
2006-07-27         NaN         NaN         NaN         NaN          NaN
2006-07-28         NaN         NaN         NaN         NaN          NaN
…                  …           …           …           …            …
2026-11-04        0.89        0.91        0.93        0.97         0.96
2026-11-05        0.93        0.92        0.94        0.97         0.96
2026-11-06        0.92        0.92        0.94        0.97         0.96
2026-11-09        0.94        0.93        0.94        0.97         0.96
2026-11-10        0.94        0.93        0.94        0.97         0.96
```

```
[5108 rows x 27 columns]
```

[15]:
```
for window in rolling_windows.keys():
    plot_time_series(
        df=rolling_corr_results_df,
        plot_start_date=None,
        plot_end_date=None,
        plot_columns=[col for col in rolling_corr_results_df.columns if␣
 ↪f"{window}" in col],
        title=f"Figure {next(figure_number)}: Rolling {window} Correlation",
        x_label="Date",
        x_format="Year",
        x_tick_spacing=1,
        x_tick_start=None,
        x_tick_rotation=30,
        y_label="Correlation",
        y_format="Decimal",
        y_format_decimal_places="Auto",
        y_tick_spacing="Auto",
        y_tick_rotation=0,
        grid=True,
        legend=True,
        export_plot=False,
        plot_file_name=None,
        caption=f"Data Source: Yahoo Finance, Analysis by the Author",
    )
```

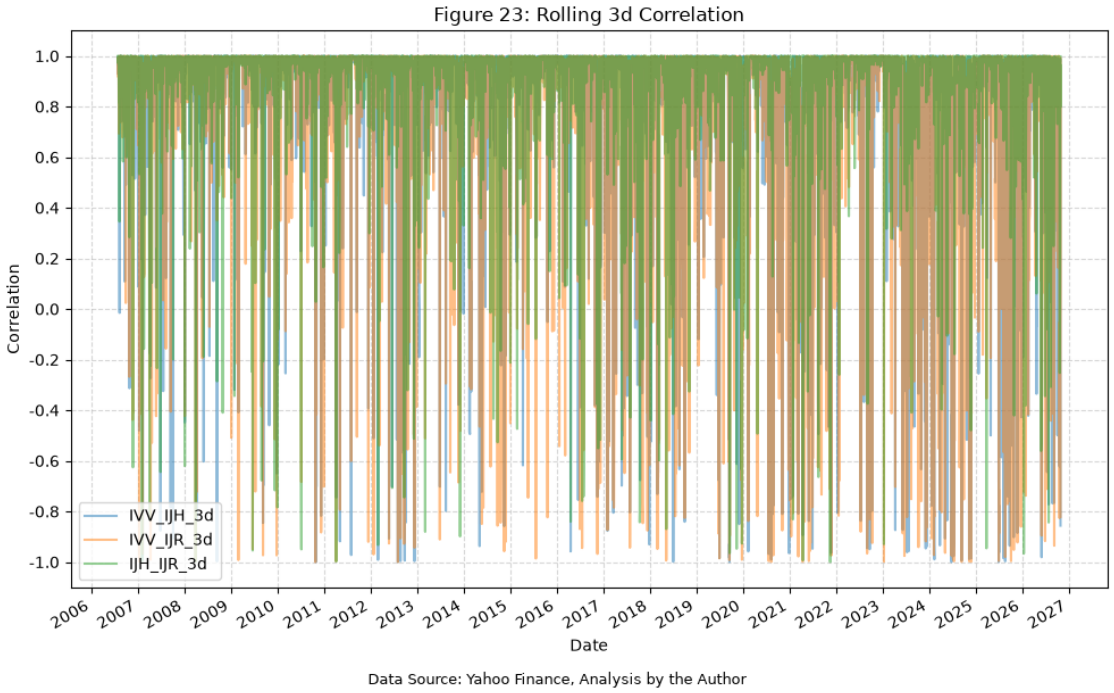
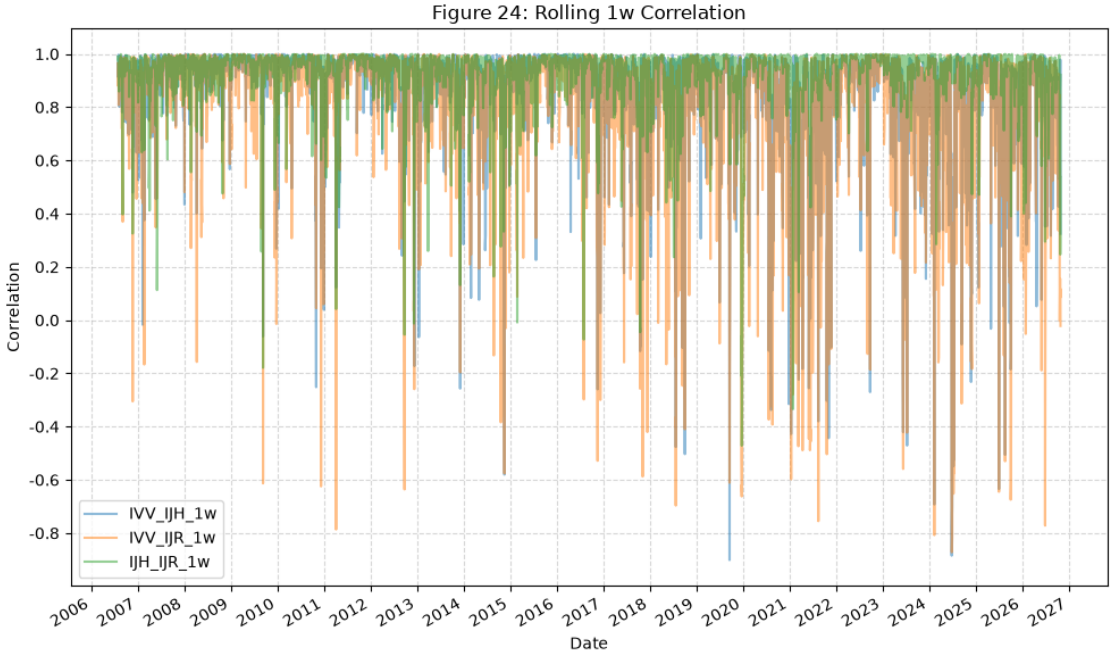
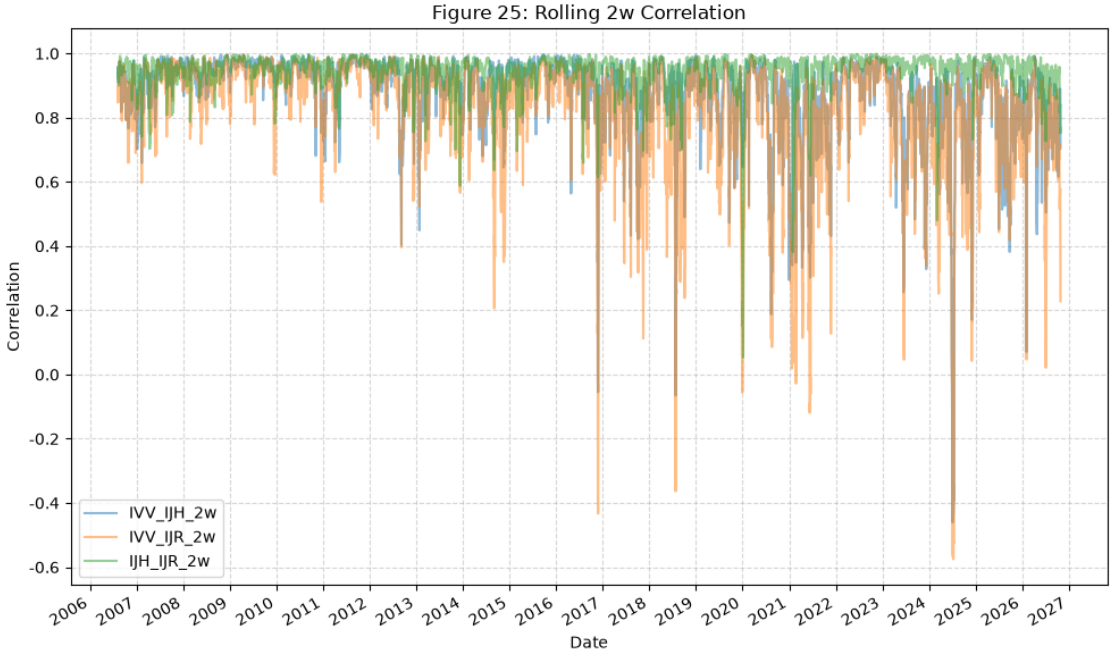


Figure 24: Rolling 1w Correlation



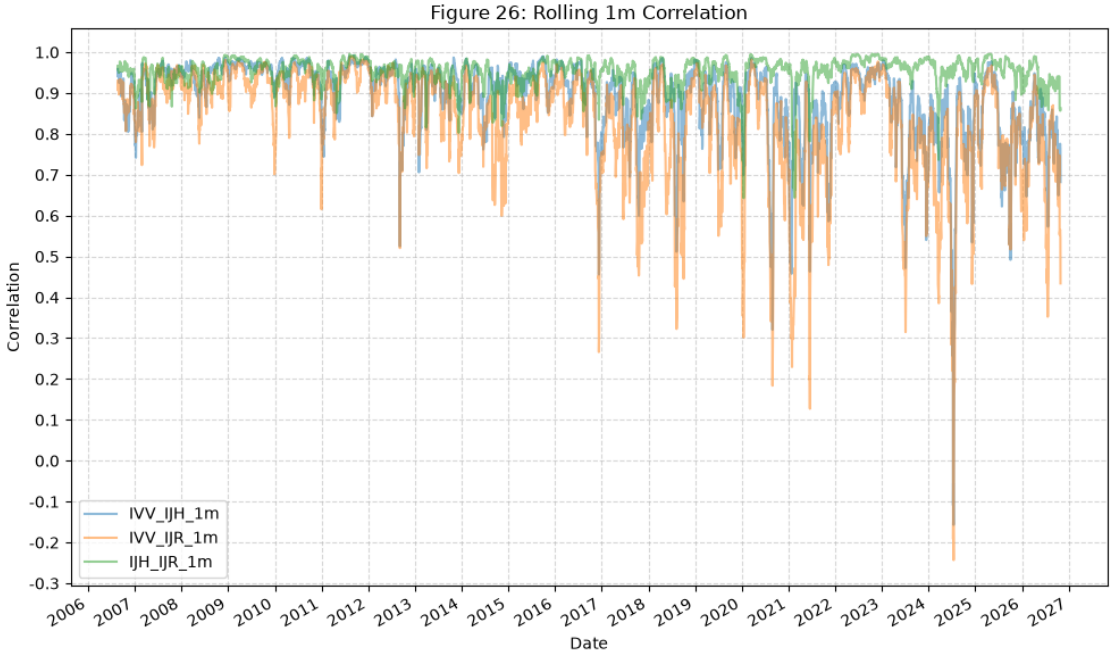
Data Source: Yahoo Finance, Analysis by the Author

Figure 25: Rolling 2w Correlation



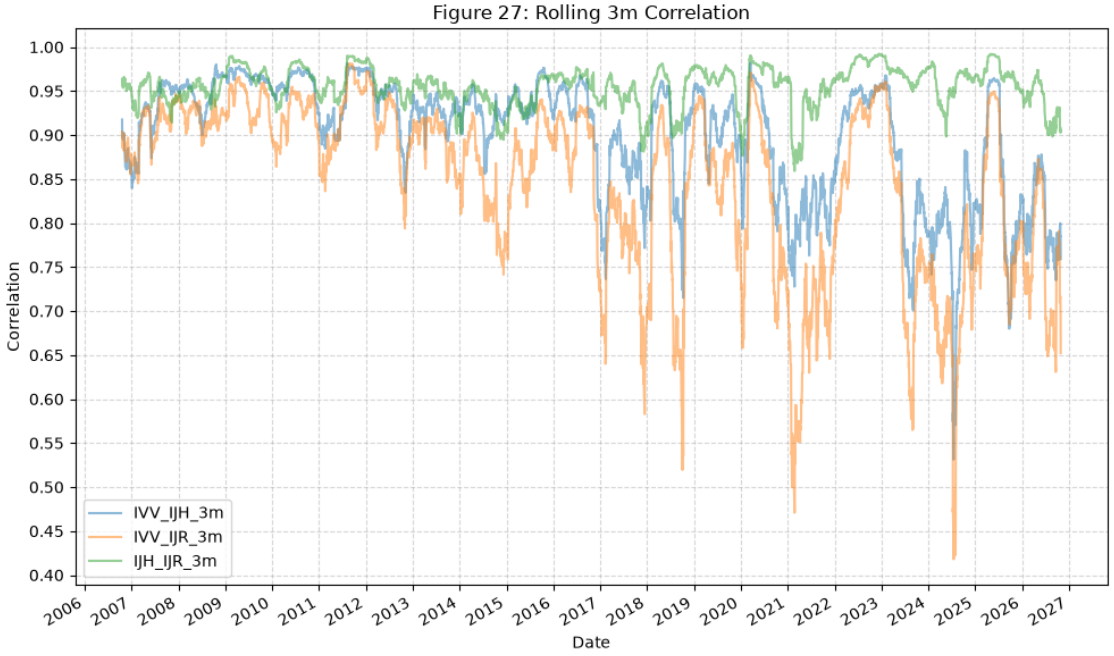
Data Source: Yahoo Finance, Analysis by the Author

Figure 26: Rolling 1m Correlation



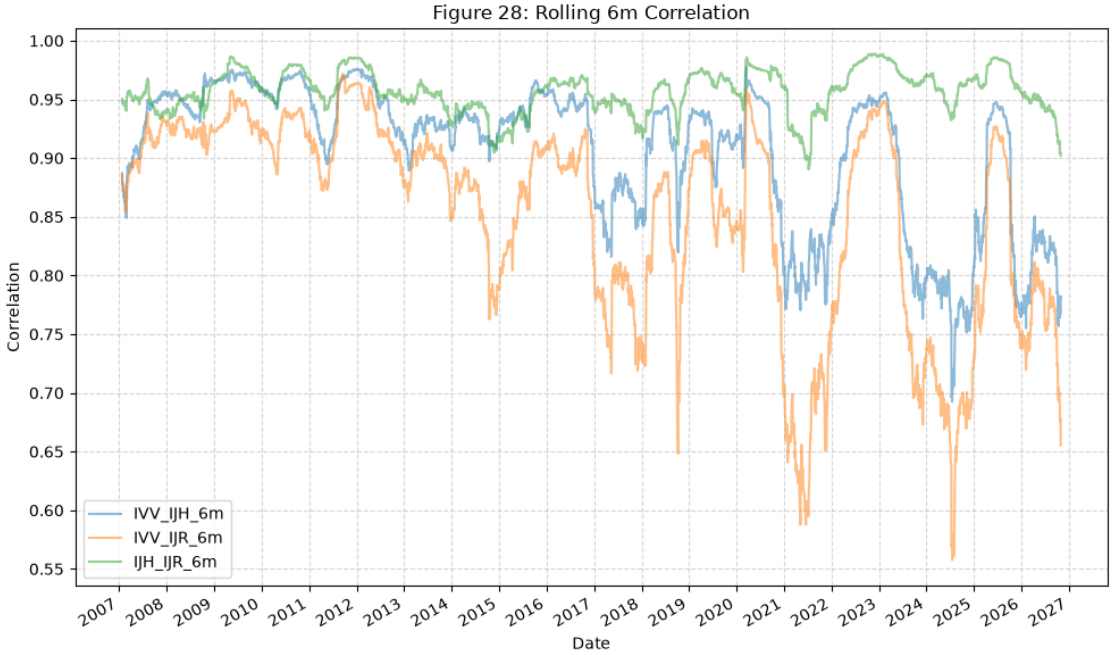
Data Source: Yahoo Finance, Analysis by the Author

Figure 27: Rolling 3m Correlation



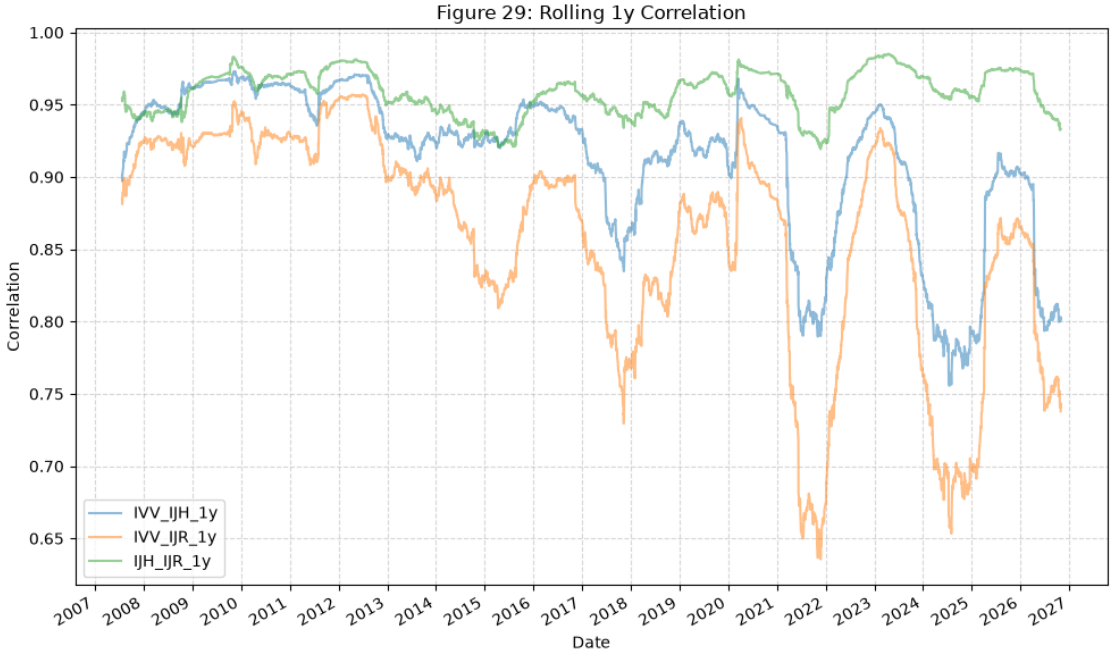
Data Source: Yahoo Finance, Analysis by the Author

Figure 28: Rolling 6m Correlation



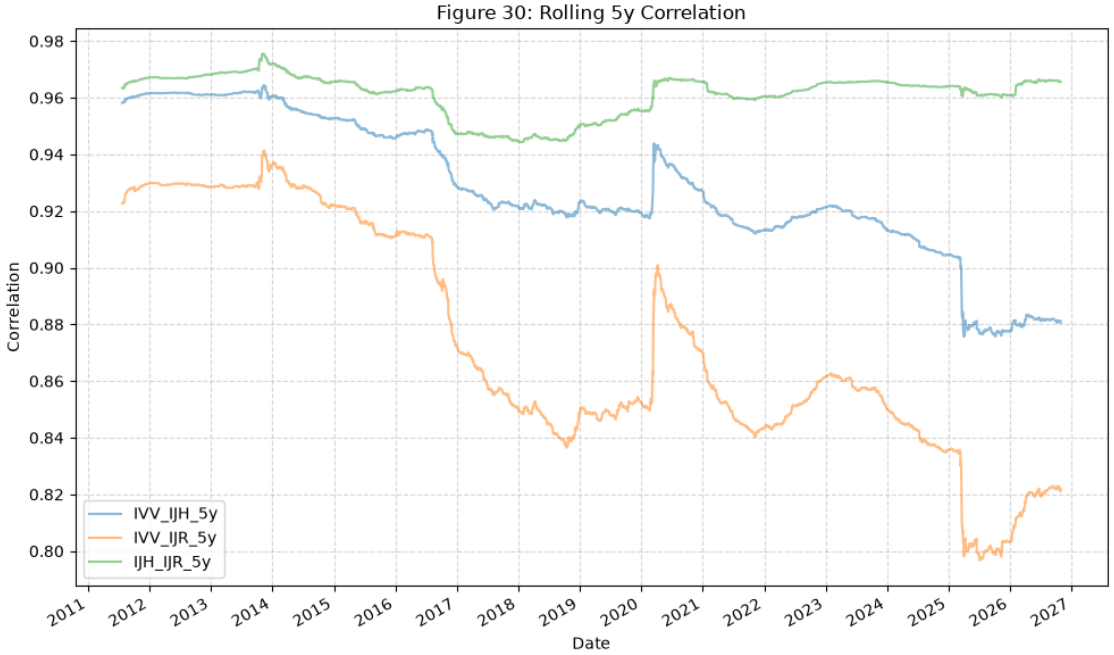
Data Source: Yahoo Finance, Analysis by the Author

Figure 29: Rolling 1y Correlation



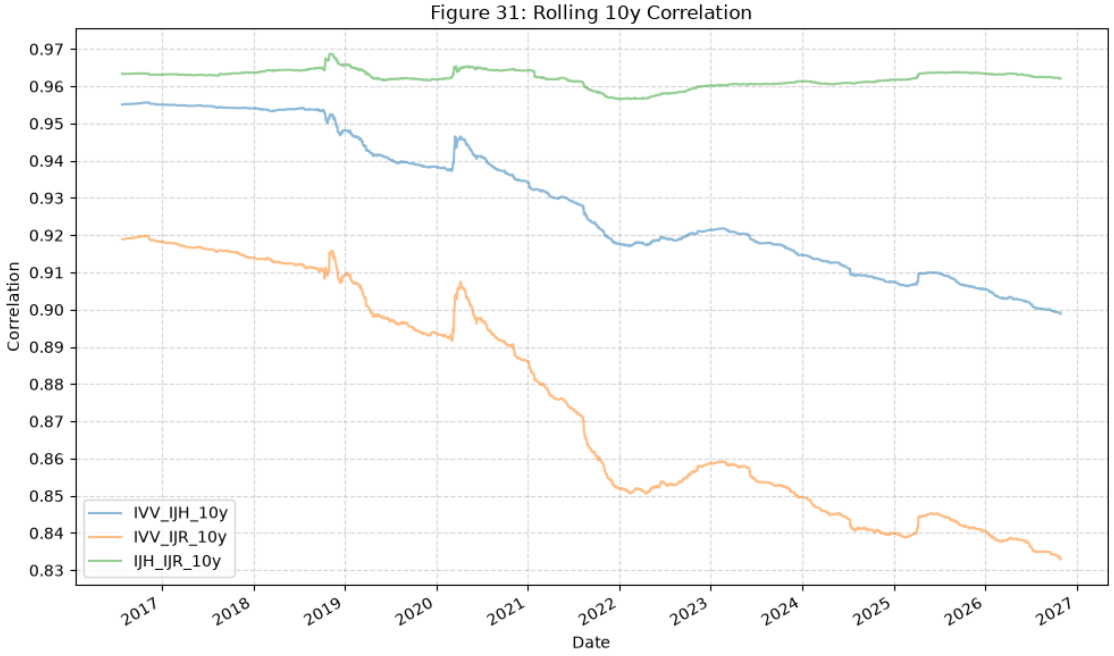
Data Source: Yahoo Finance, Analysis by the Author

Figure 30: Rolling 5y Correlation



Data Source: Yahoo Finance, Analysis by the Author

Figure 31: Rolling 10y Correlation



Data Source: Yahoo Finance, Analysis by the Author

With the above plots, we can see that there are periods of time when the correlations abruptly increase or decrease, such as:

- Early-mid 2007 (the start of the financial crisis)
- Mid 2011 (the European debt crisis)
- Late 2015 into 2016
- Late 2018 (rate hikes?)
- Early 2020 (COVID-19 pandemic)
- Late 2022 into early 2023 (rate hikes, recession fears, COVID tech bubble, etc.)
- Mid 2024 (banking crisis, rate hikes)
- Early 2025 (Liberation day)

We're not necessarily looking to explain away each of these time periods or delve into the macro factors that may have been at play or driving the correlations, but we are more interested in the change in correlation over time as a response to some kind of market shock or event.

0.7 Conclusion

Overall, this was a brief but interesting study that dove into the historical correlations amongst a wide range of asset classes. The full period correlation matrices gave us the expected picture, while the rolling correlations showed how much those relationships change over time, particularly in response to market shocks.

0.8 Future Investigation

In part 2, we'll investigate whether the rolling correlations can be used as a signal, by building a "total correlation" metric, looking at future returns based on that metric, and developing a trading strategy.

0.9 References

None